

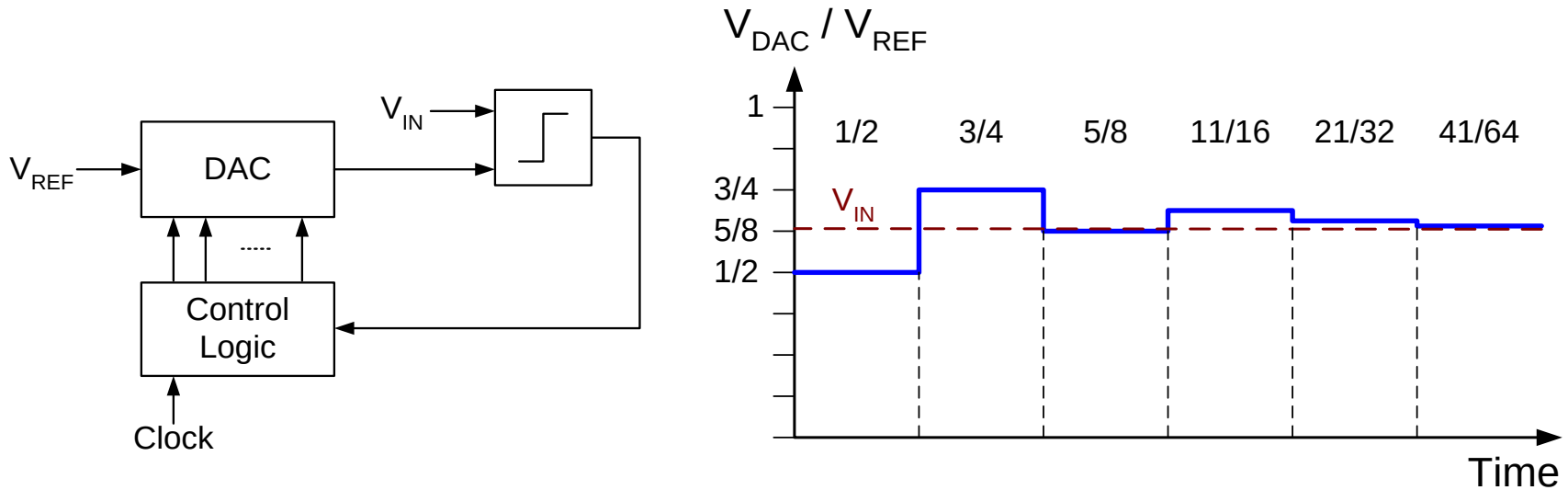
Low-Power SAR ADC Design

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Content

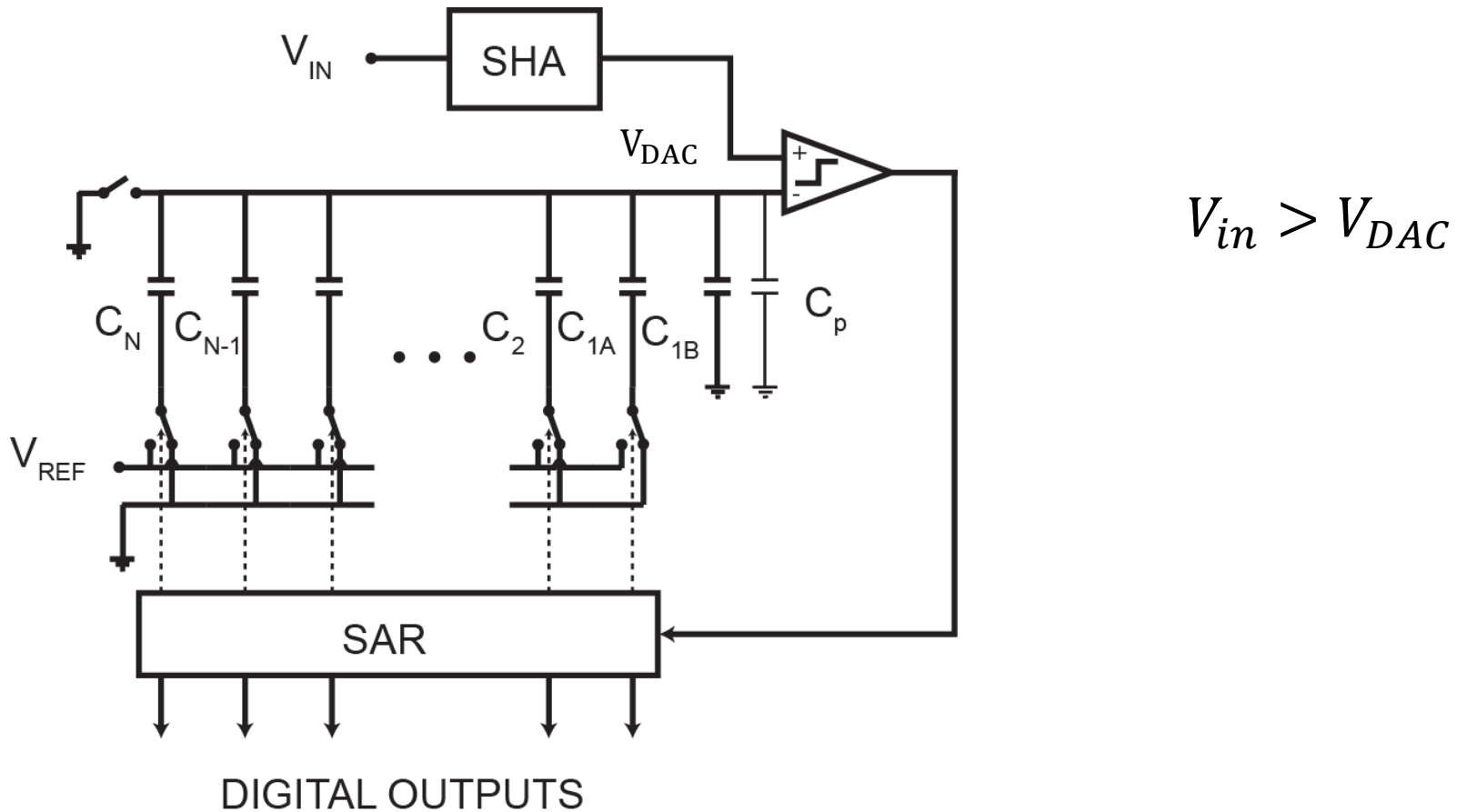
- Basic operation of SAR
- SAR ADC noise analysis
- Comparator noise reduction techniques
- Low power DAC switching techniques

SAR ADC



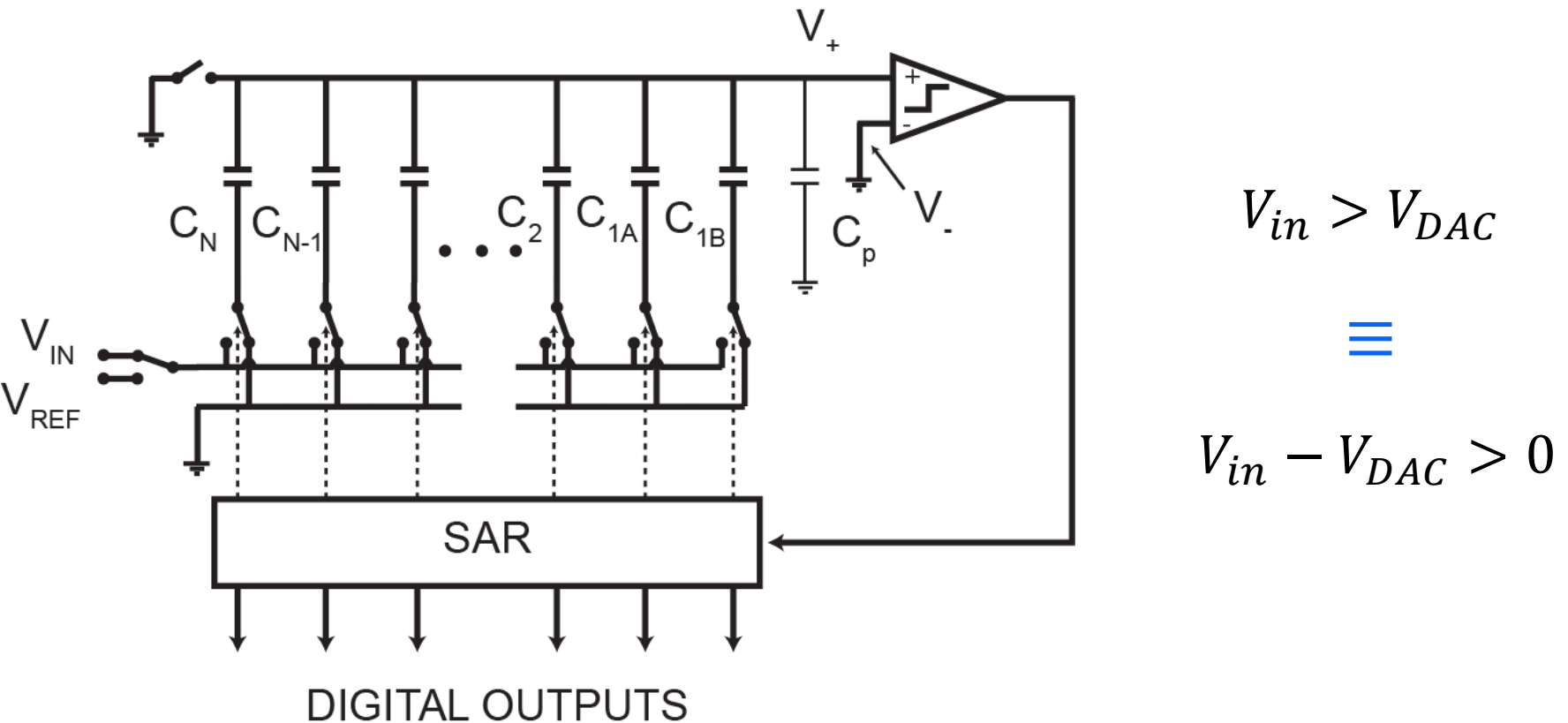
- Binary search over DAC output
- High accuracy achievable
 - Rely on highly accurate comparator
- Moderate speed

SAR ADC Direct Implementation



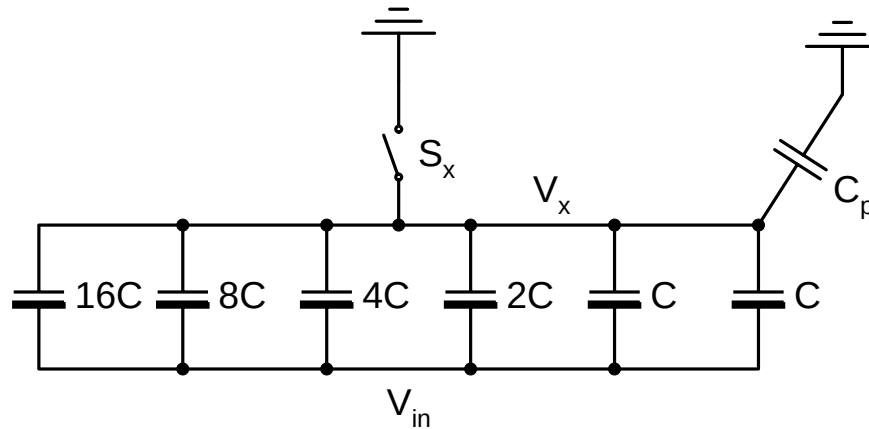
- Comparator common-mode variation $\rightarrow$ Signal dependent offset
- Nonlinear capacitor $C_p \rightarrow$ Cause gain error and INL

Classic Implementation



- Reuse the SAR DAC for input sampling; inherent subtraction
- Comparator common-mode voltage at Gnd
- Nonlinear parasitic capacitance C_p has no effect

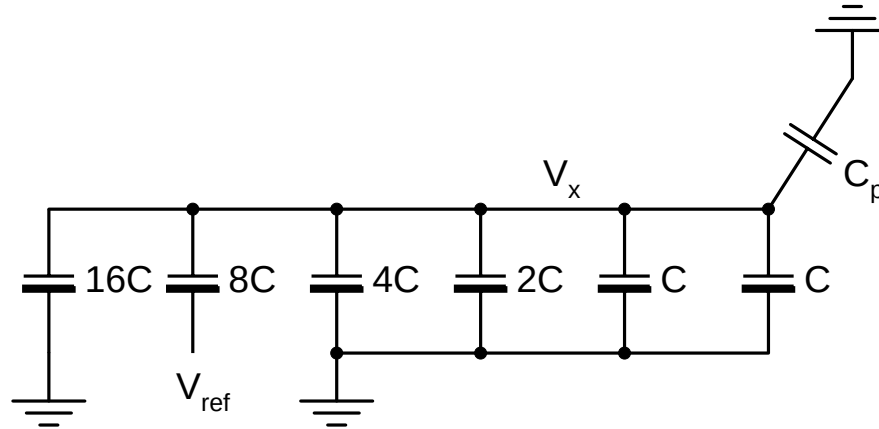
Sampling Phase (5-bit Example)



- Total charge at node V_x after opening S_x

$$Q = -V_{in} \cdot 32C = -V_{in} \cdot C_{total}$$

Bit4 Test (Assuming Bit5=0)



$$Q = -V_{in} \cdot C_{total} = (V_x - V_{ref}) \cdot 8C + V_x \cdot (24C + C_p)$$

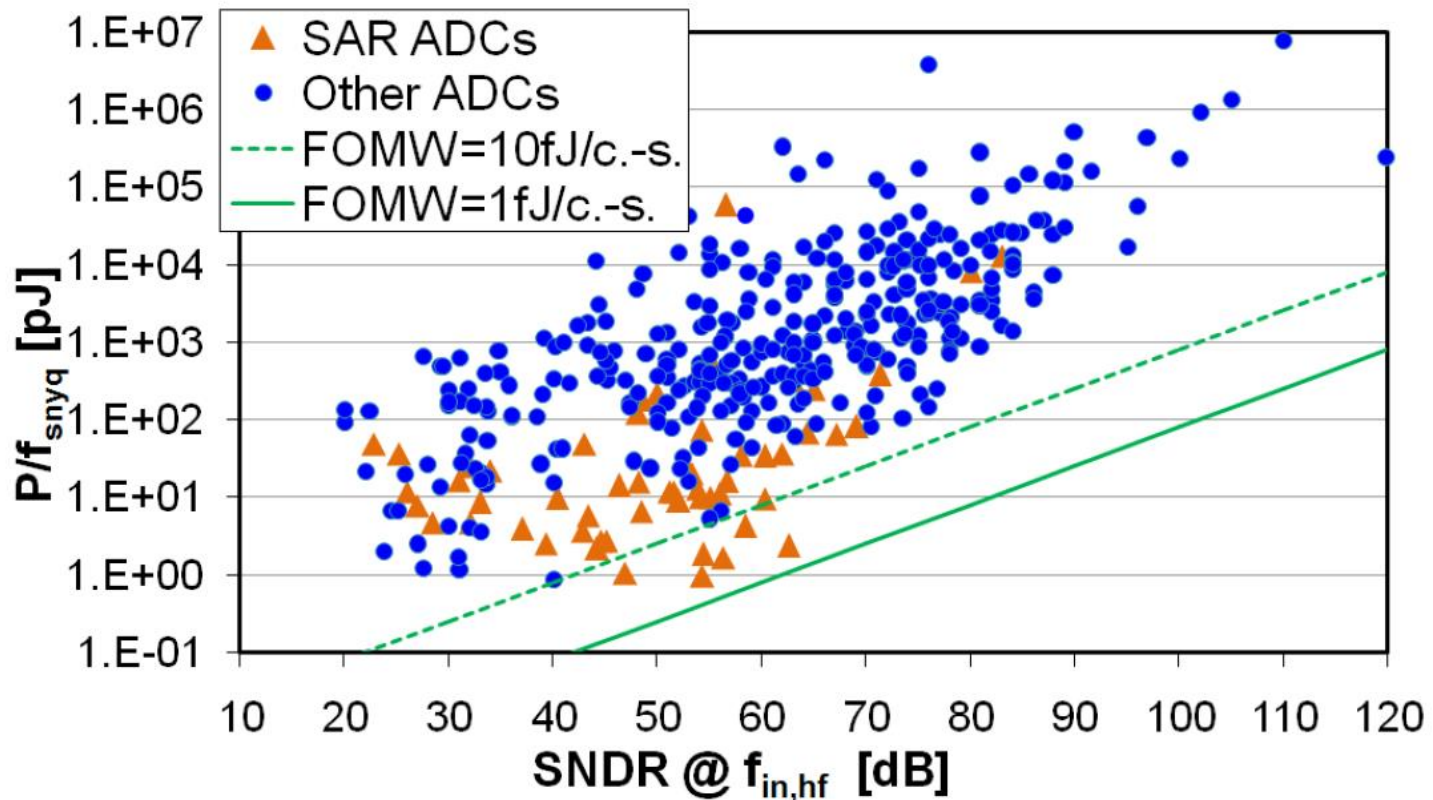
$$\therefore V_x = \left(\frac{1}{4} V_{ref} - V_{in} \right) \cdot \frac{C_{total}}{C_{total} + C_p}$$

- $V_x < 0 \Rightarrow V_{in} > 0.25V_{ref} \Rightarrow \text{Bit4}=1$
- $V_x > 0 \Rightarrow V_{in} < 0.25V_{ref} \Rightarrow \text{Bit4}=0$

Advantages

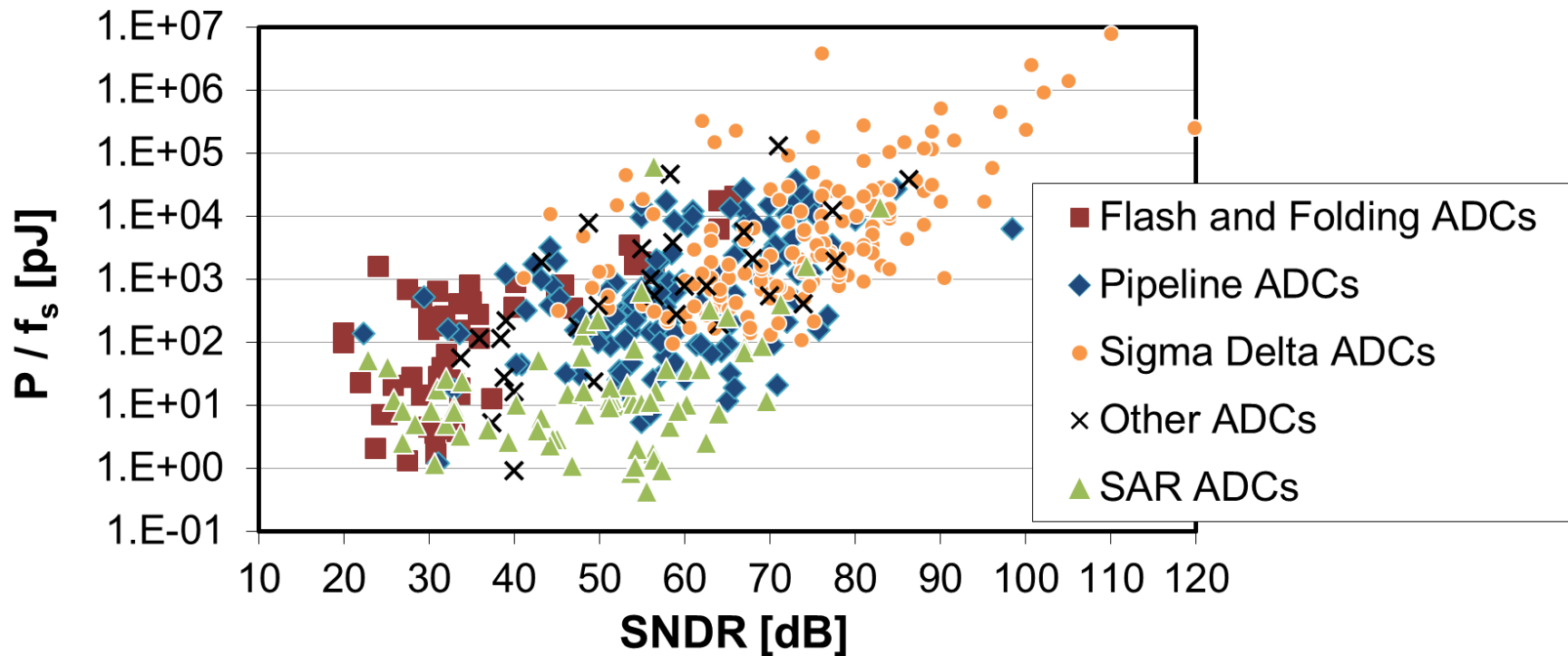
- SAR ADC is simple
 - A DAC, a comparator, a SAR logic
 - Short design cycle, fast time to market
- SAR ADC is scaling friendly
 - Mostly digital
 - No static power
 - Low area and low cost

SAR ADCs are highly energy efficient



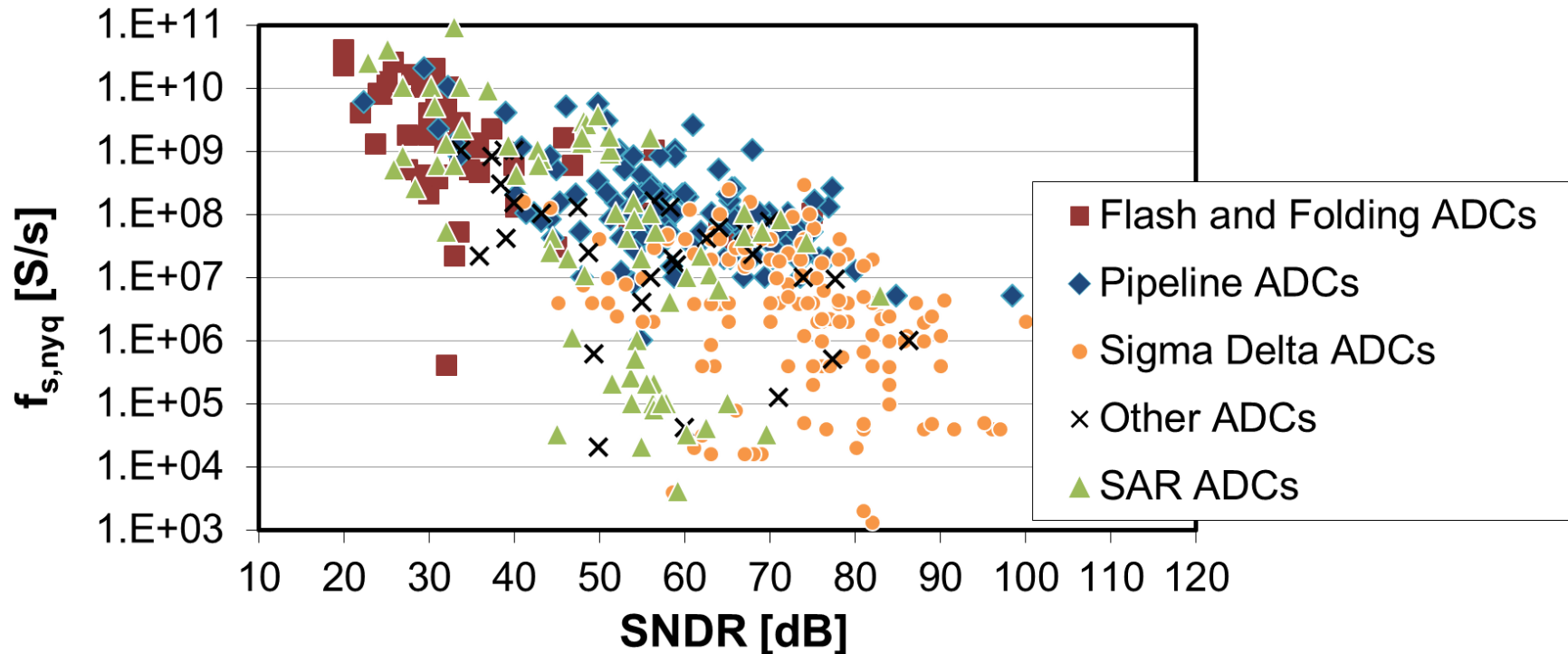
B. Murmann, "ADC Performance Survey 1997-2013. "
<http://www.stanford.edu/~murmann/adcsurvey.html>.

ADC Architectures: Power Efficiency



- SAR ADCs are highly power efficient

ADC Architectures: Speed vs Accuracy



- SAR ADCs are becoming fast

Limitations

- Not well suited for high resolution (10-bit)
 - Require large capacitor
 - Problematic for input and reference driving
 - Large DAC switching power
 - Require low-noise comparator
 - Require large power consumption
- Conversion rate is limited
 - Require N conversion cycles for N -bit
 - DAC RC settling, comparator delay, SAR logic delay

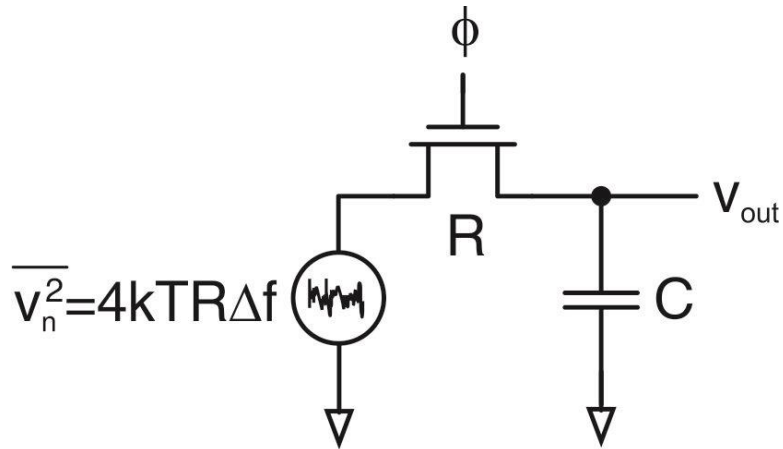
Content

- Basic operation
- Low-noise and low-power design techniques
- High-speed design techniques
- Low-area design techniques
- DAC mismatch calibration

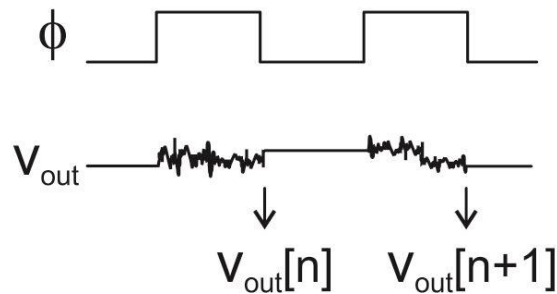
SAR ADC noise analysis

- Noise in the sampling process
 - kT/C noise (part of the discrete-time input after sampling)
- Noise in the conversion process
 - DAC noise
 - Comparator noise

kT/C noise

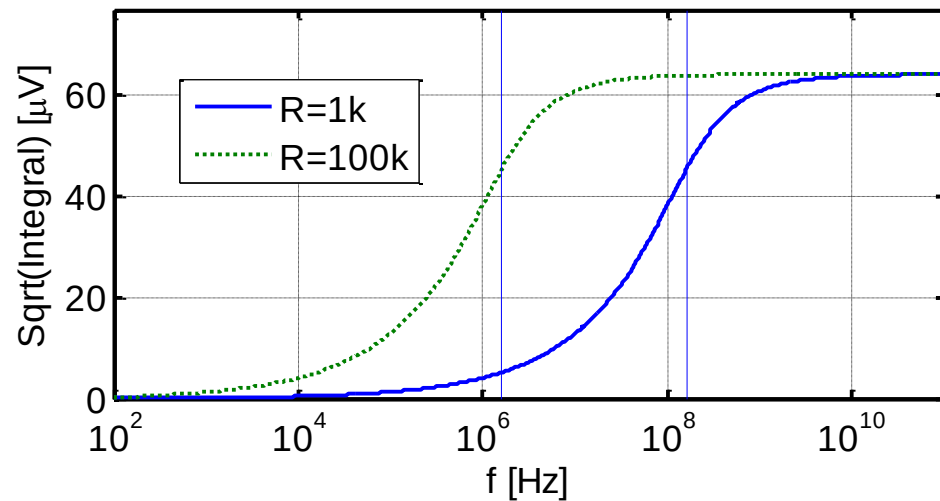
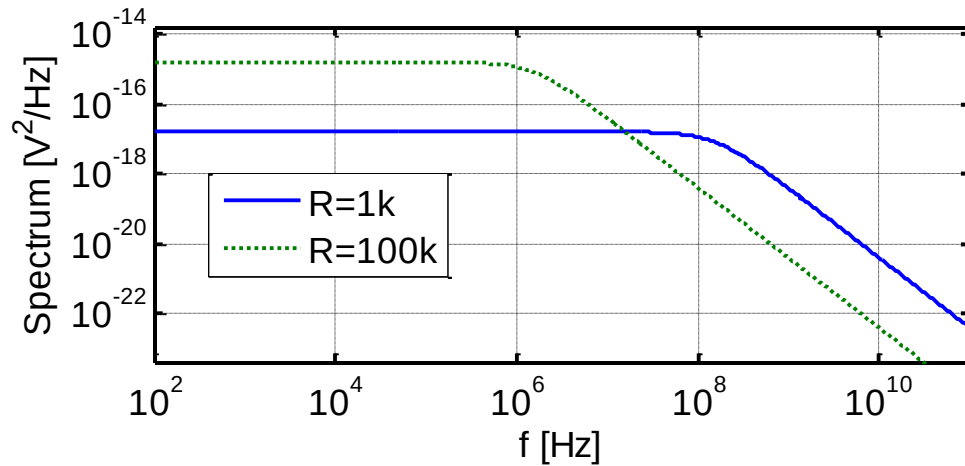


$$\frac{\overline{v_{out}^2}}{\Delta f} = 4kTR \left| \frac{1}{1 + sRC} \right|^2$$



$$\overline{v_{out}^2[n]} = 4kTR \int_0^\infty \frac{1}{1 + (2\pi fRC)^2} df = \frac{kT}{C}$$

Impact of R



Increasing R increases the noise PSD, but also decreases the BW

Equipartition Theorem

- Equipartition theorem in statistical physics states:
 - In thermal equilibrium each degree of freedom holds on average thermal energy of $kT/2$
- In an electronic circuit, each independent capacitor holds an electric energy of $kT/2$ and each independent inductor holds a magnetic energy of $kT/2$.
- This theorem can help reduce the complexity of noise analysis
 - But it does not apply to active circuits that are not in equilibrium

$$\frac{1}{2} C \overline{v_{out}^2} = \frac{1}{2} kT$$
$$\Rightarrow \overline{v_{out}^2} = \frac{kT}{C}$$

kT/C Noise

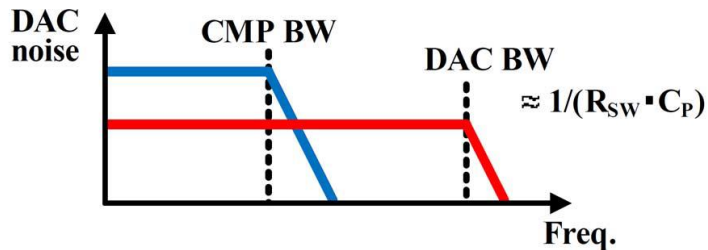
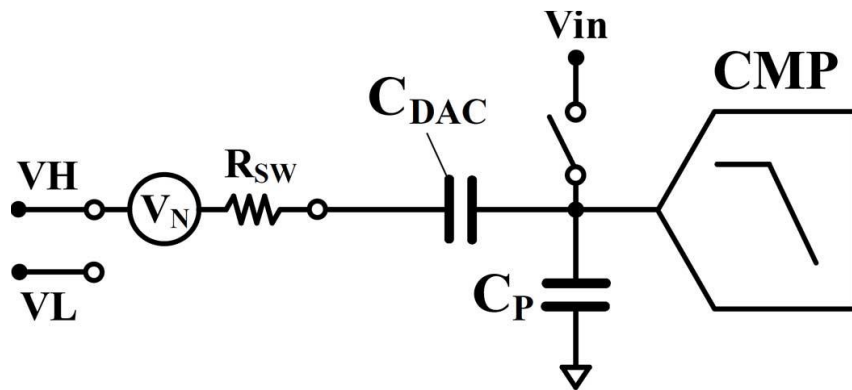
- Assuming the kT/C noise equal to the quantization noise of a B-bit ADC

$$\frac{kT}{C} = \frac{\Delta^2}{12} \Rightarrow C = 12kT \cdot \left(\frac{2^B}{V_{FS}} \right)^2$$

- Assuming $V_{FS} = 1V$:

B	C [pF]
8	0.003
10	0.052
12	0.834
14	13.3
16	213
18	3416

DAC noise



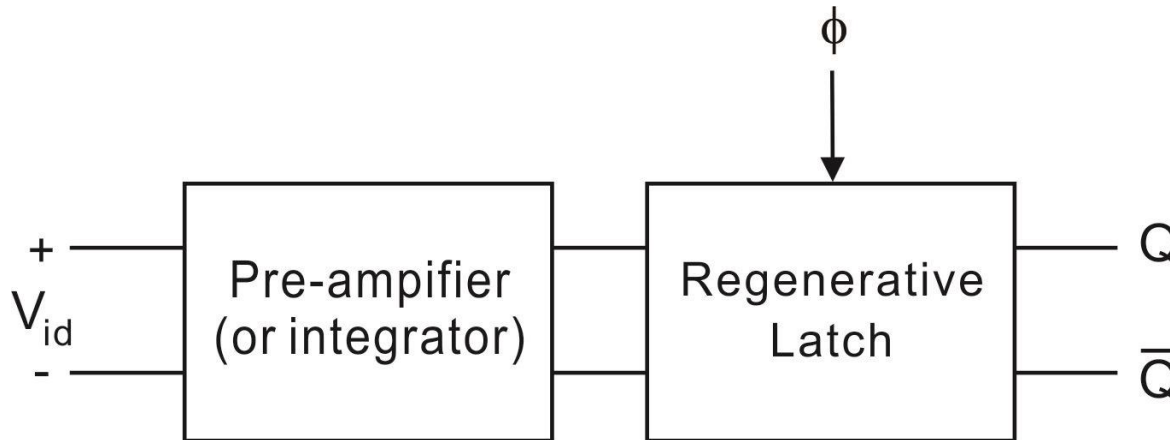
- DAC noise PSD is set by R_{SW}
- DAC settling BW is set by C_{DAC}
- DAC noise BW is set by C_P
 - $C_{DAC} \gg C_P$
- Total integrated DAC noise is kT/C_P , which is large, but not all of it affects the comparator decision
- For DAC BW $\gg$ comparator BW, the majority of the DAC noise is filtered out
 - Comparator BW $\sim$ DAC settling BW
- Best way to suppress DAC noise is to make R_{SW} small
 - Consume low power
 - Also speed up DAC settling
 - Benefit from CMOS scaling

Comparator noise

- Comparator noise analysis
- Circuit-level low-noise techniques
- Architectural-level low-noise techniques

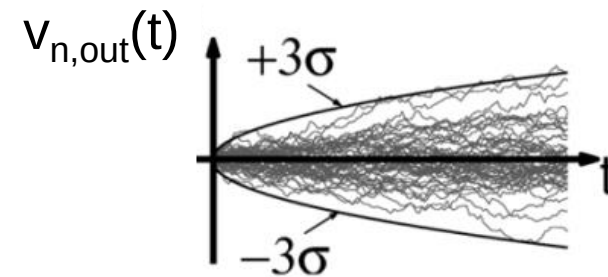
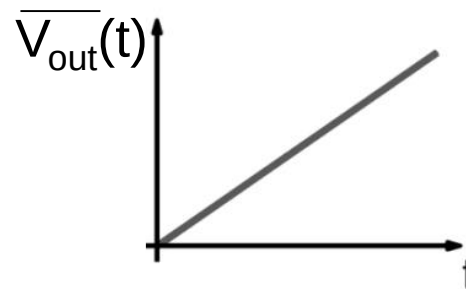
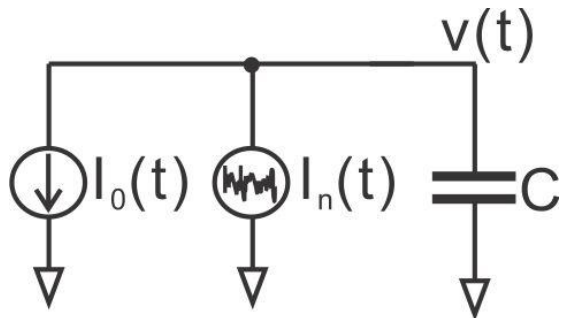
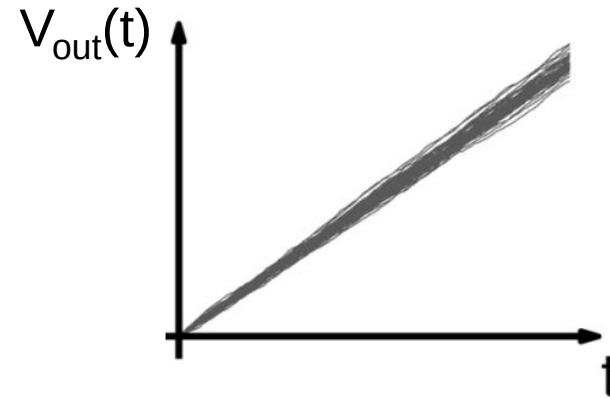
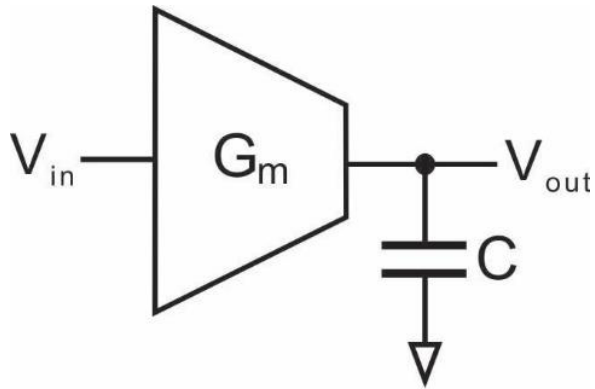
Comparator

- Standard comparator configuration
 - 1st-stage dominates noise and offset
 - The noise analysis for a static pre-amp is straightforward

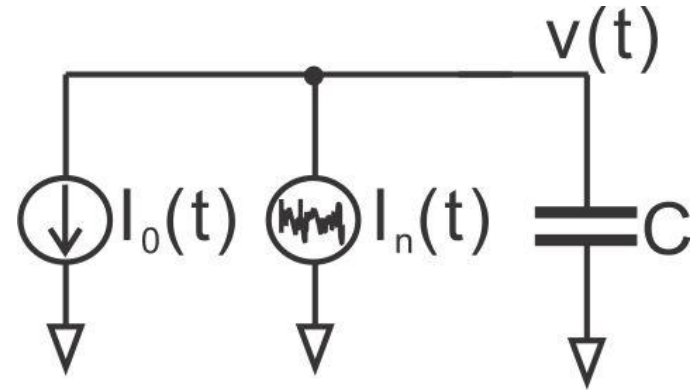
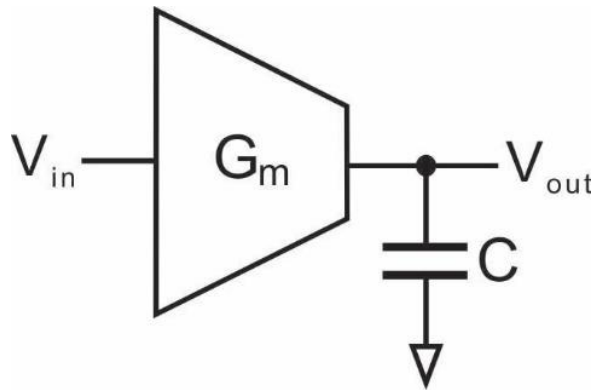


$$\sigma_n = \sqrt{\sigma_{n,preamp}^2 + \frac{\sigma_{n,latch}^2}{G^2}}$$

Noise analysis for a dynamic integrator

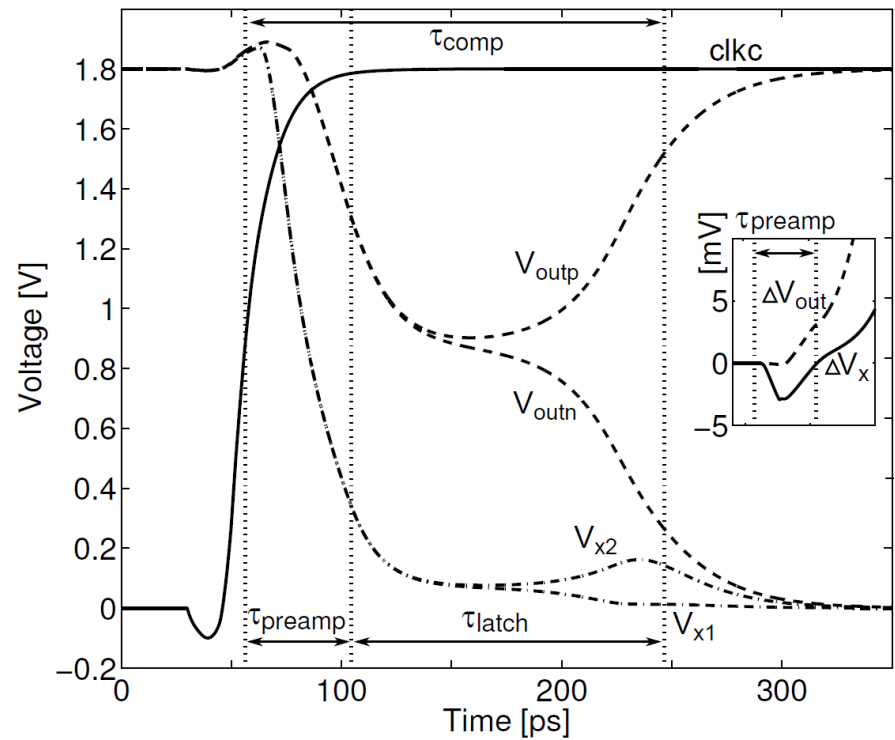
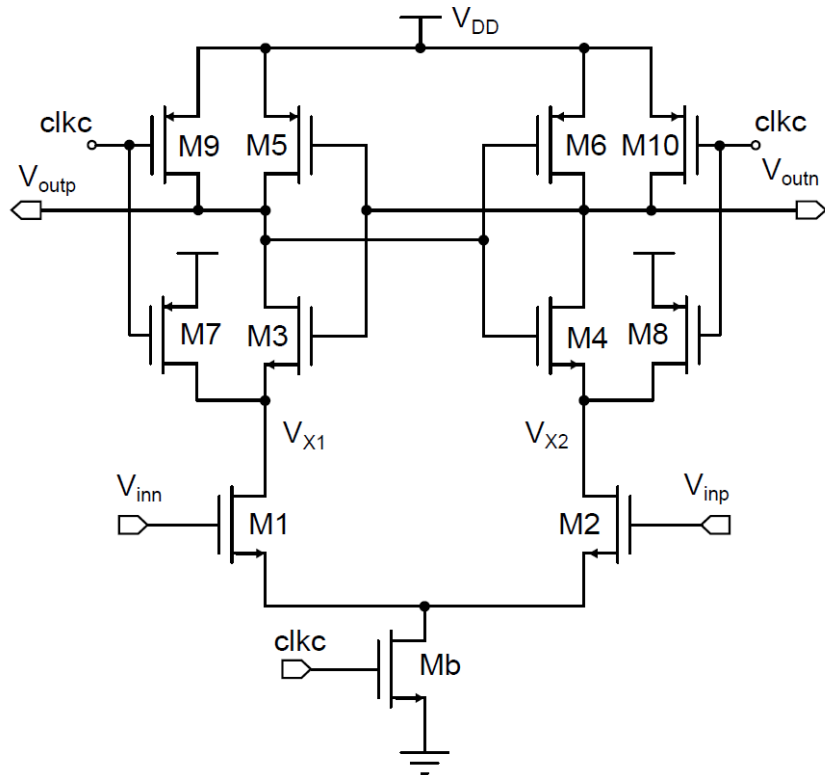


Noise analysis for a dynamic integrator

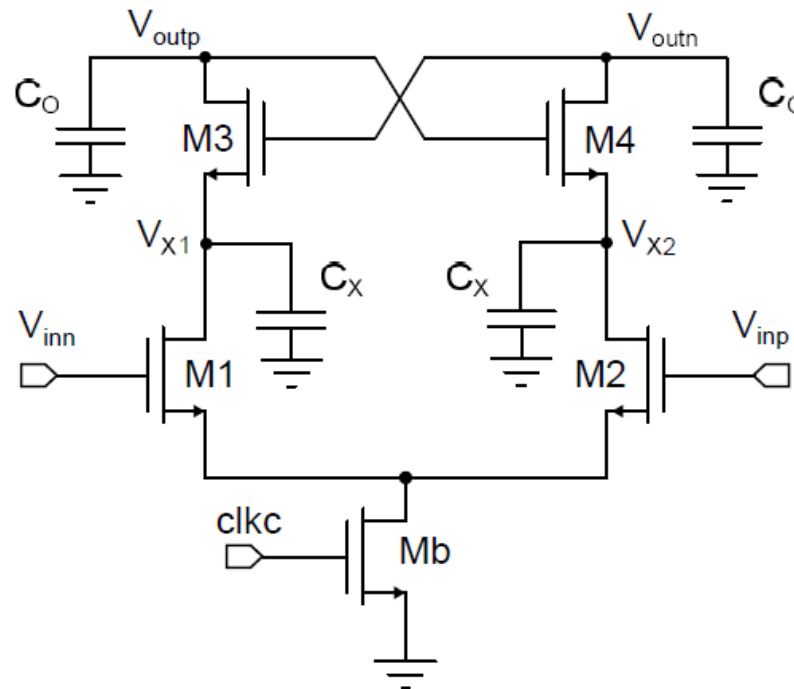


- $G = \frac{v_{out}}{v_{in}} = \frac{g_m T_{int}}{C}$
- $E = P \cdot T_{int} = V_{dd} \cdot I_D \cdot T_{int}$
- $\overline{v_{n,out}^2} = \frac{kT}{C} + \frac{2kT\gamma g_m T}{C^2} = \frac{kT}{C} (1 + 2\gamma G) \approx \frac{2kT}{C} \cdot \gamma G$
- $\overline{v_{n,in}^2} = \frac{\overline{v_{out}^2}}{G^2} \approx \frac{2kT}{C} \cdot \frac{\gamma}{G} = \frac{2kT\gamma}{g_m T_{int}} = \frac{4kT\gamma}{g_m} \cdot \underbrace{\frac{1}{2T_{int}}}_{\text{Noise BW}}$
- $\underbrace{\overline{v_{n,in}^2} \cdot E}_{\text{Noise energy product}} = 2kT\gamma \frac{V_{dd}}{g_m/I_D}$

Strong-Arm Latch

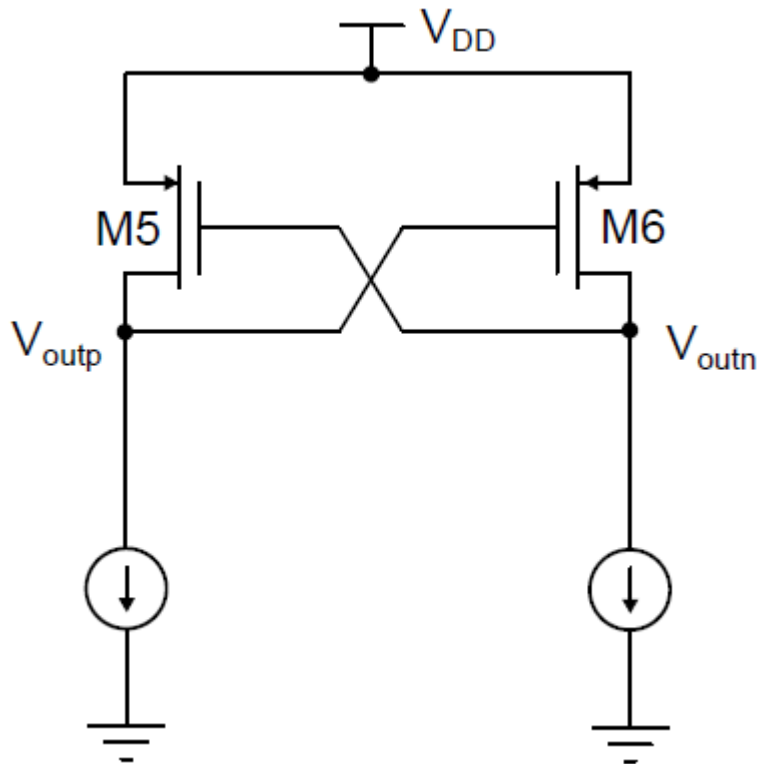


Pre-amplification phase



- $T_{int} = Q_{int}/I_D = [C_x(V_{tn} + V_{tp}) + C_o V_{tp}]/I_D$
- $G = \frac{g_m T_{int}}{C_o} = \frac{g_m}{I_D} \cdot \frac{C_x(V_{tn} + V_{tp}) + C_o V_{tp}}{C_o}$
- $\overline{\sigma_{n,preamp}^2} = \underbrace{2}_{\text{Differential}} \cdot \frac{2kT\gamma}{g_m/I_D} \cdot \frac{1}{C_x(V_{tn} + V_{tp}) + C_o V_{tp}}$

Latch phase



- Both signal and noise grows exponentially
 - Input referred noise does not decrease with time
- $\overline{\sigma_{n,latch}^2} = \frac{2(1+\gamma)kT}{C_o}$
- More comprehensive analysis can be found from [P. Nuzzo, TCAS-I, 2008]

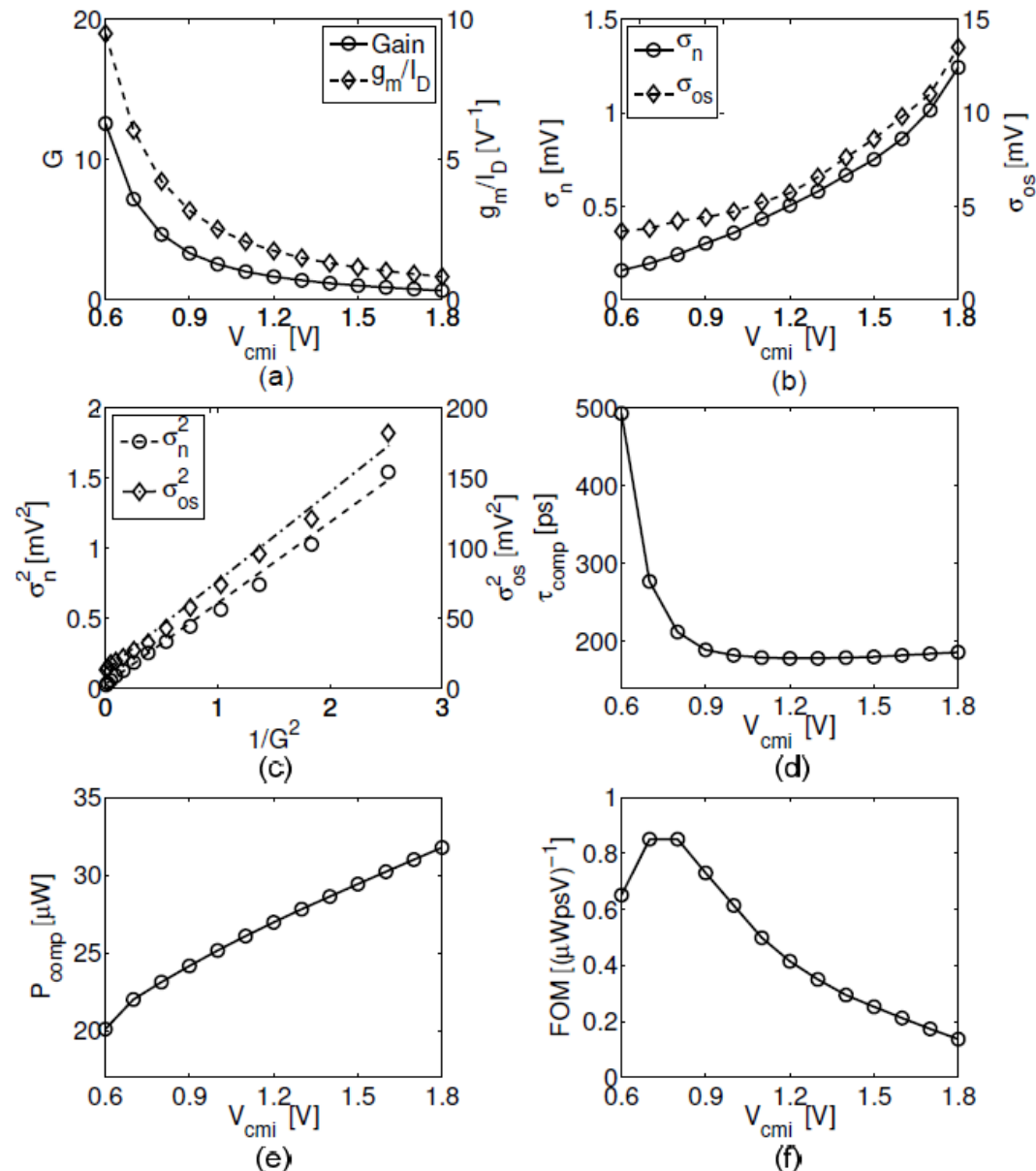
Comparator noise analysis

- For a well-optimized comparator, we prefer:
 - Large G : attenuate the noise from the latch stage
 - In practice, G is around 5
 - A too-large G slows down the comparator
 - Regeneration is faster than integration
 - Large g_m/I_D : it improves energy efficiency
 - In practice, g_m/I_D is around 15 V^{-1}
 - A too-large g_m/I_D slows down the comparator; it also increases the comparator input capacitance, which causes signal attenuation and leads to increase ADC input referred noise
- Offset analysis is very similar to noise

$$\sigma_{os} = \sqrt{\sigma_{os,preamp}^2 + \frac{\sigma_{os,latch}^2}{G^2}}$$

Design trade-off regarding V_{GS} (V_{cmi})

- A smaller V_{GS} increases g_m/I_d and G , but it slows down the comparator
- Comparator offset and noise increases as G decreases
- Comparator power increases gradually as V_{GS} increases
- $V_{GS} \sim V_{dd}/2$ is a good tradeoff



Two-stage comparator v1

- High-speed operation
- M10/11 starts from linear region $\rightarrow$ low gain
- Require clk and clk_bar

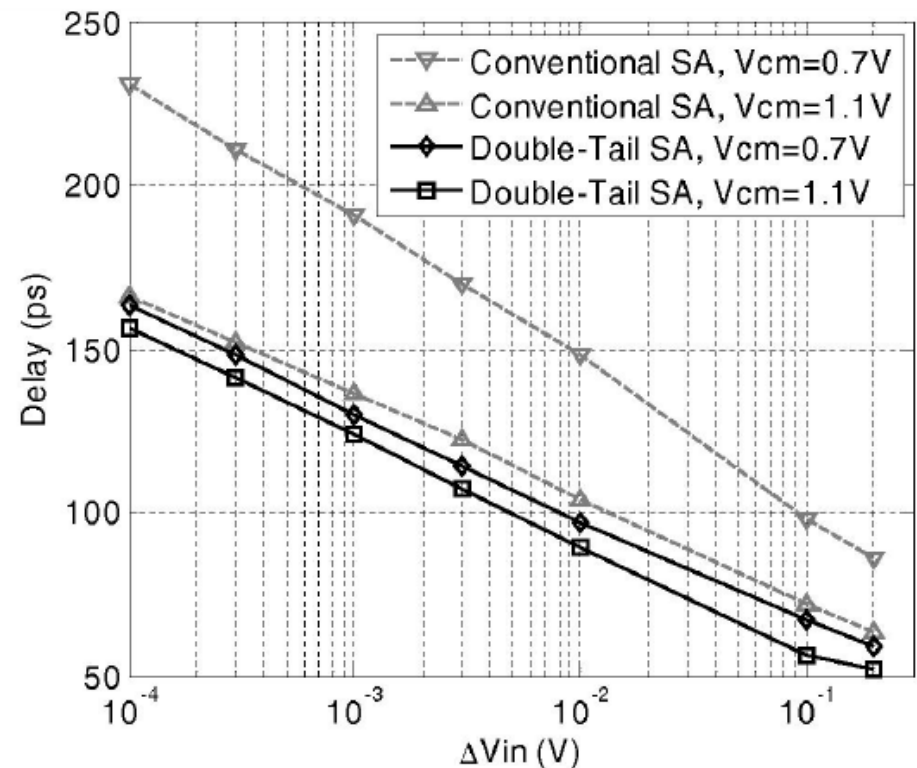
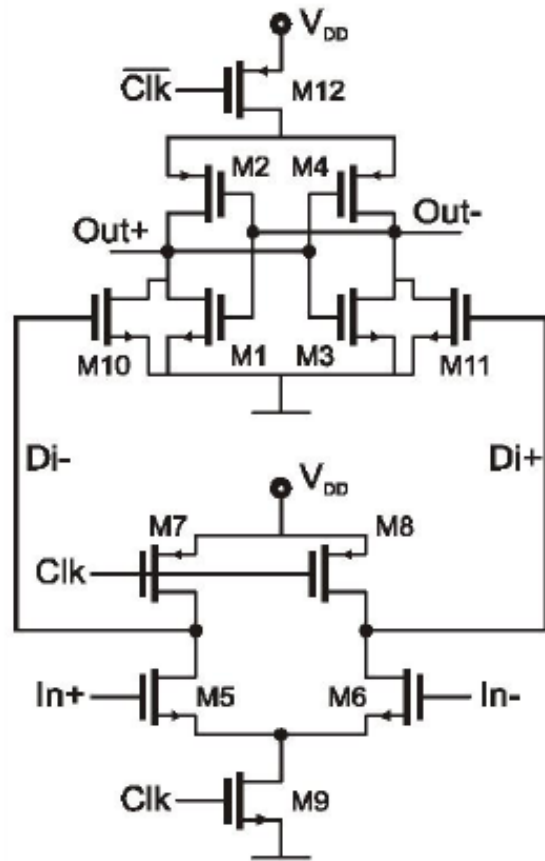
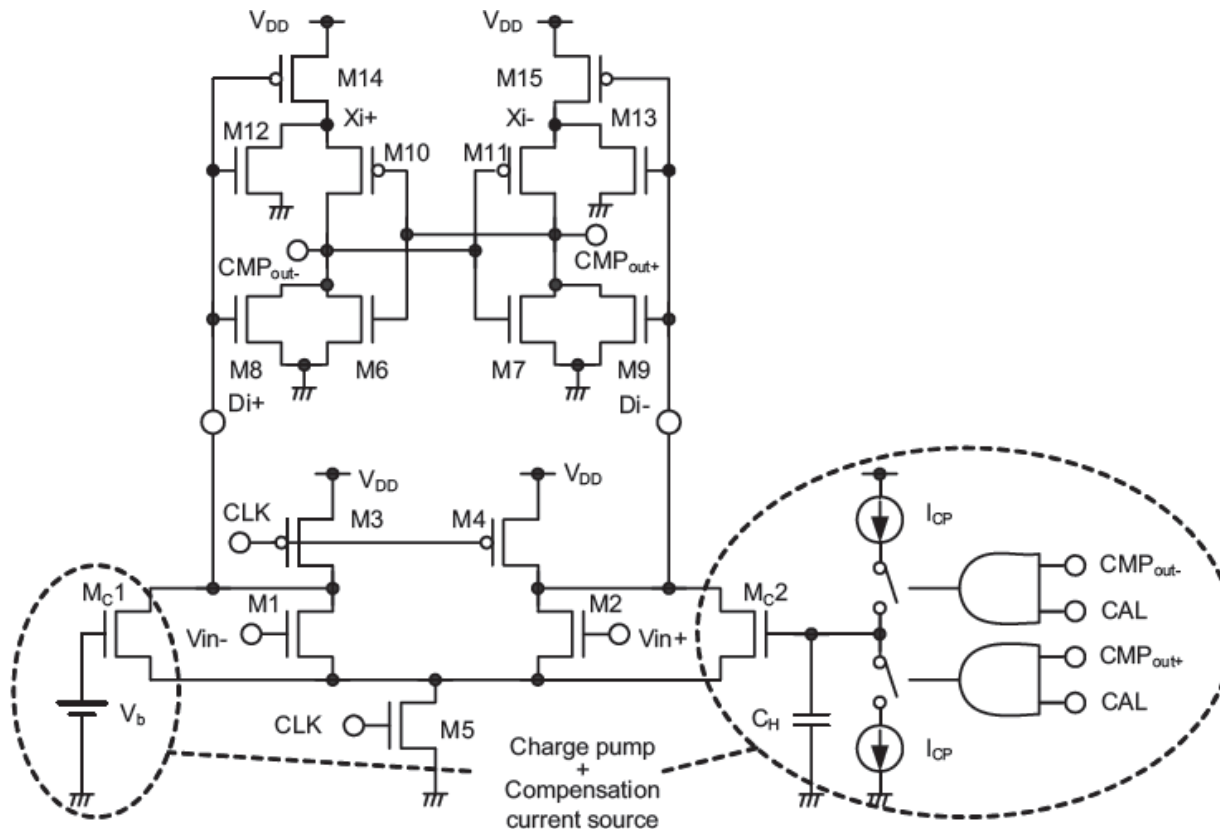


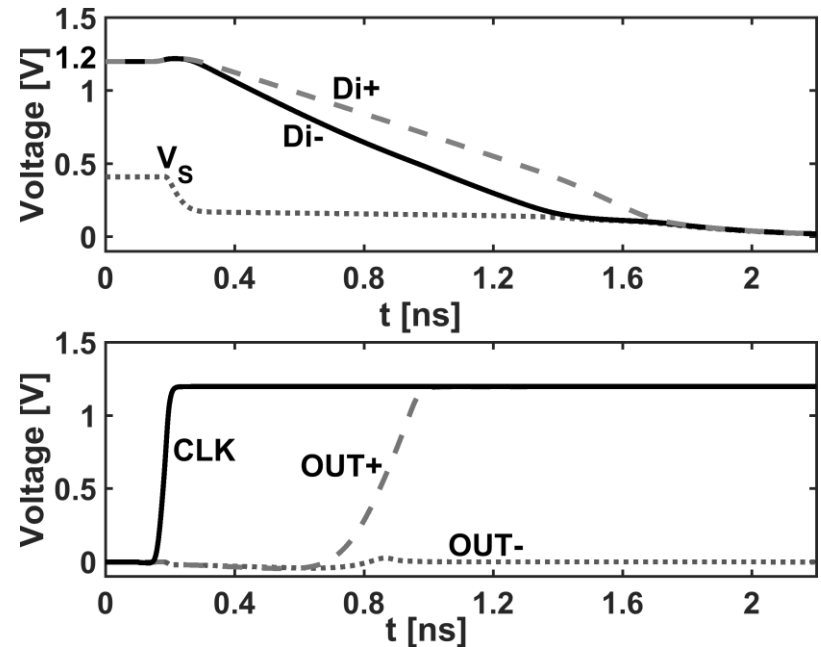
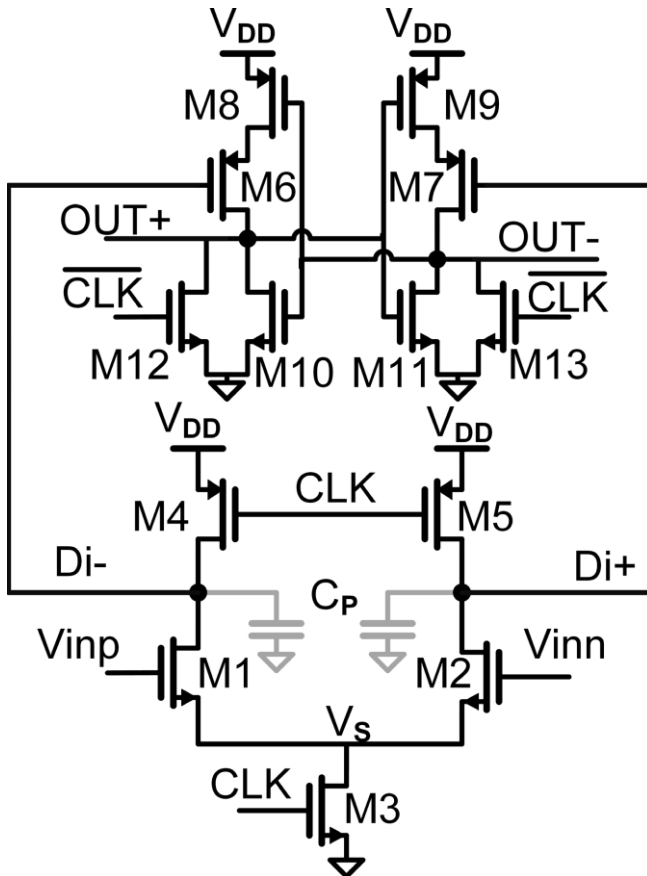
Figure 17.7.3: Simulated sense amplifier delays versus differential input voltage. The delay is the time between the clock edge and the instant when ΔOut crosses $1/2 V_{DD}$.

Two-stage comparator v2

- M8/9 starts from the saturation region $\rightarrow$ high gain
- M14 and M15 also serve as input transistors, which provide extra gain.
- Require only clk

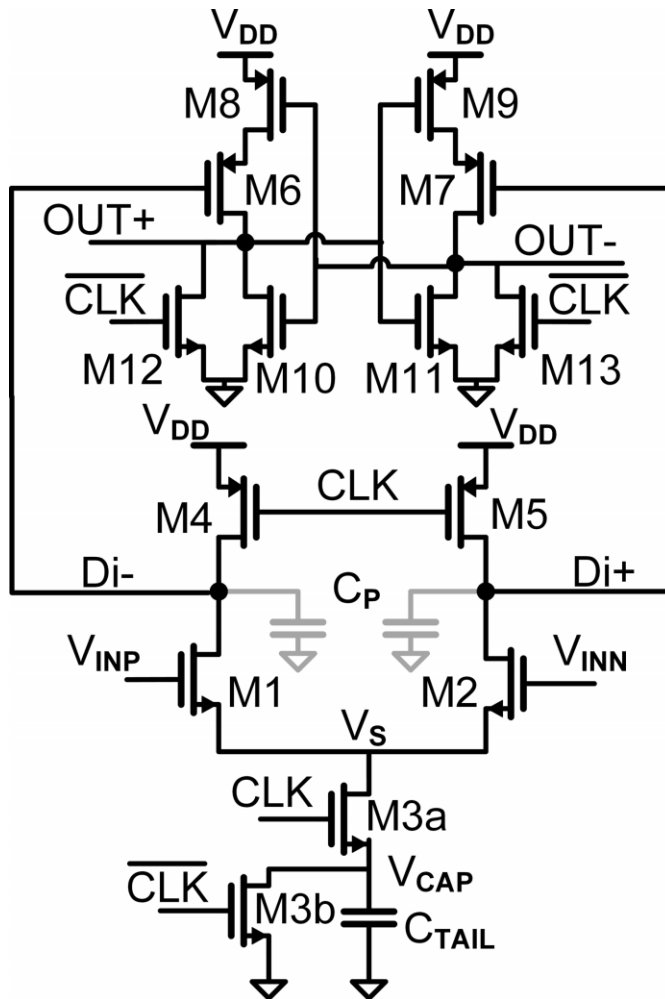


Two-stage comparator v3

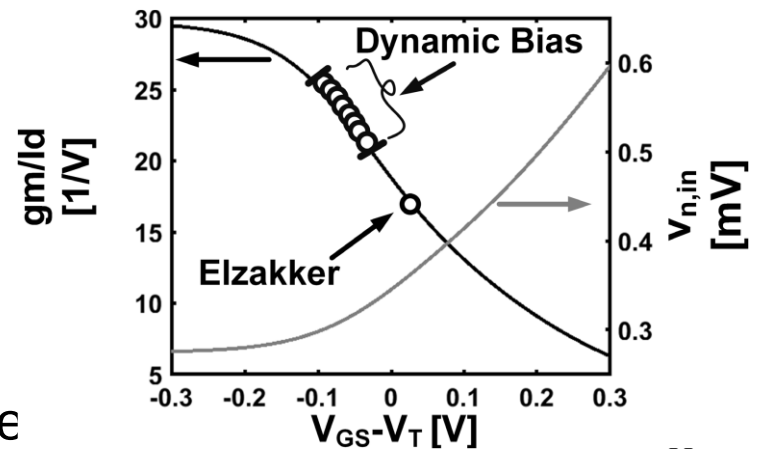
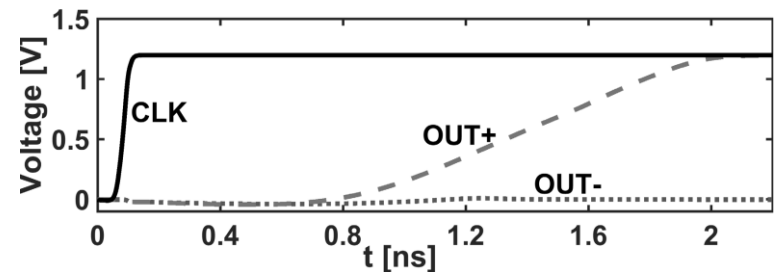
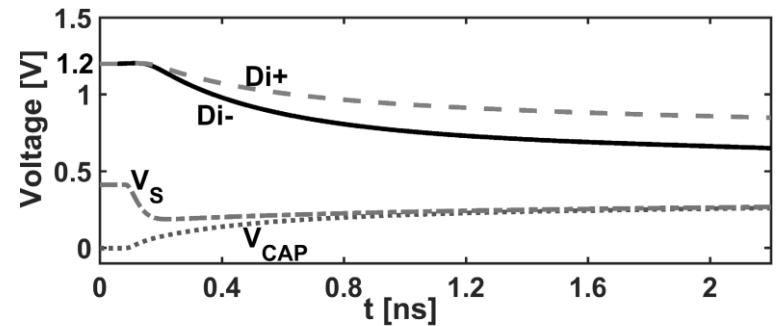


- $M6/7$ starts from the saturation region $\rightarrow$ high gain
- Require both clk and clk_bar

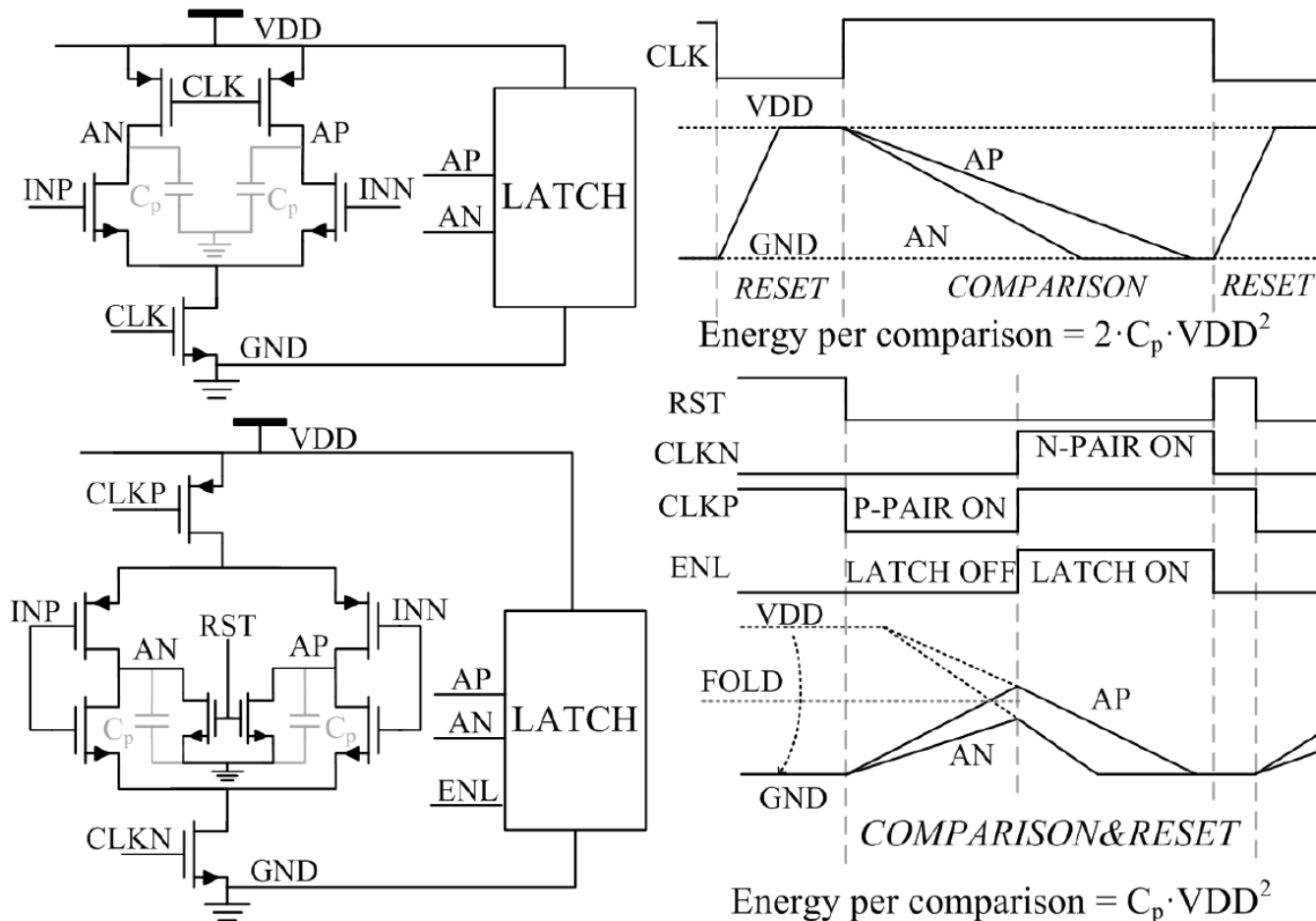
Dynamic biasing



- g_m/I_D increases as time goes by
- Output is only partially discharge



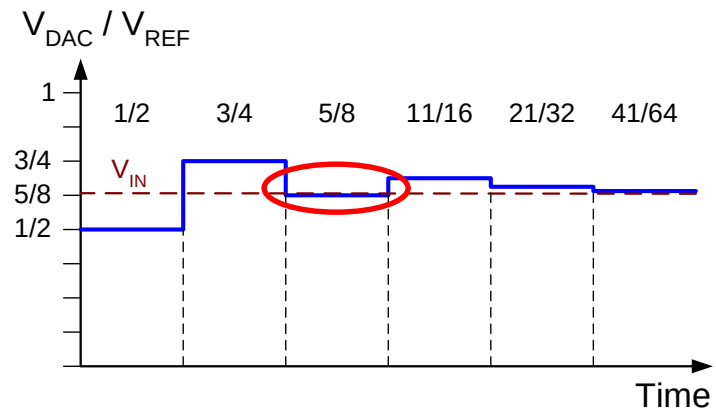
Bidirectional integration



- Power reduced by 2X due to current reuse
- 30% energy saving considering other overhead

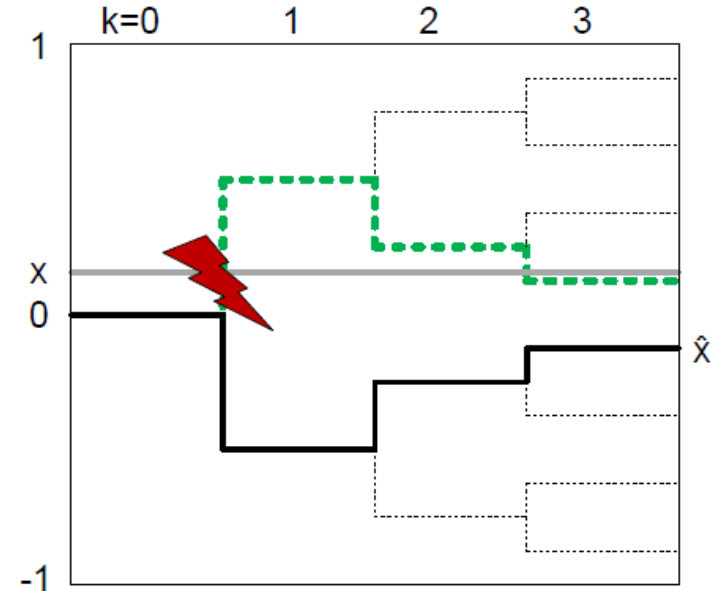
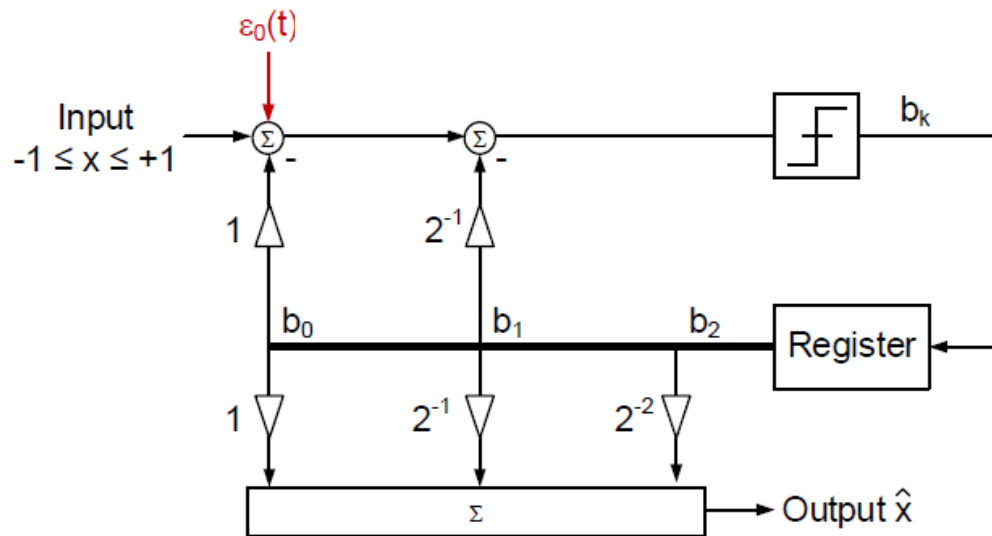
Comparator in a SAR ADC

- Architectural level properties that we can make use of
 - During entire SAR operation, there is only 1 time with comparator input less than $\Delta/2$
 - Opportunity: no need for low noise all the time
 - Challenge: we do not know when



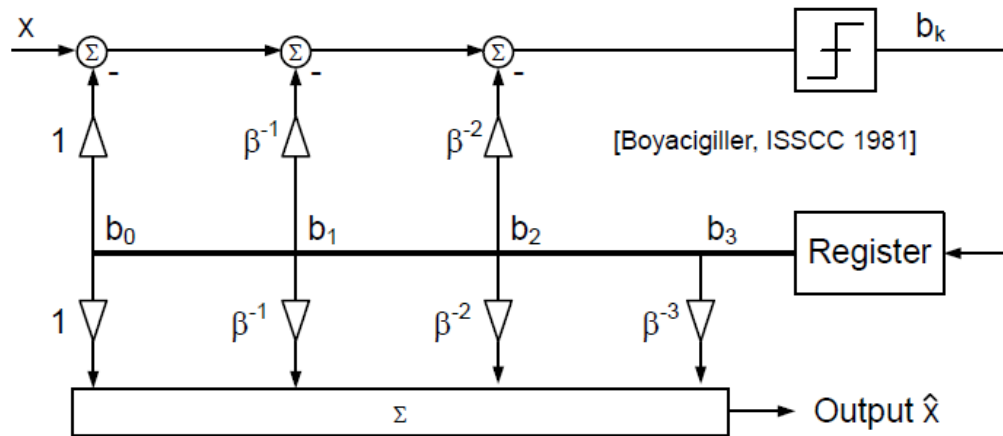
- Strategies:
 - Add redundancy at the LSB
 - Even if MSB comparison results are wrong due to high noise, we will be able to recover
 - We only need to make sure that the last 2 LSB comparisons need to be low-noise
 - Identify the noise critical comparison cycle

Concept of redundancy – no redundancy

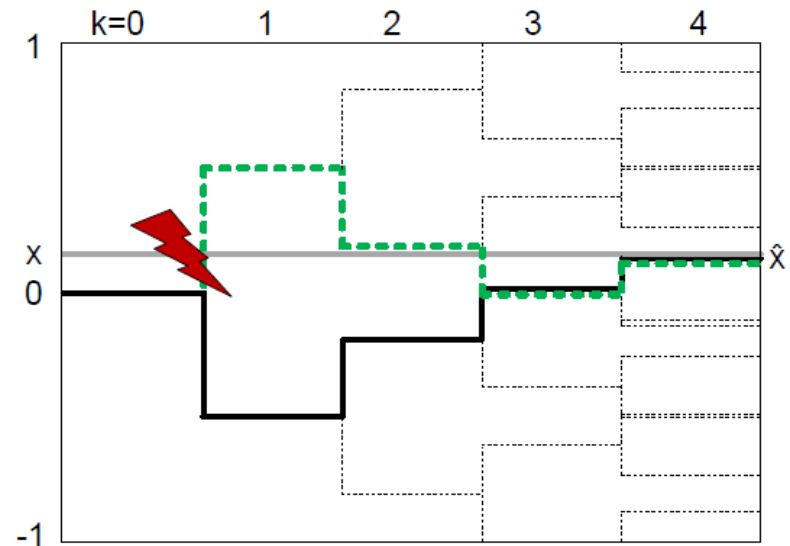


- N comparison for N bit
- Every comparison to be fully accurate

With redundancy (radix < 2)

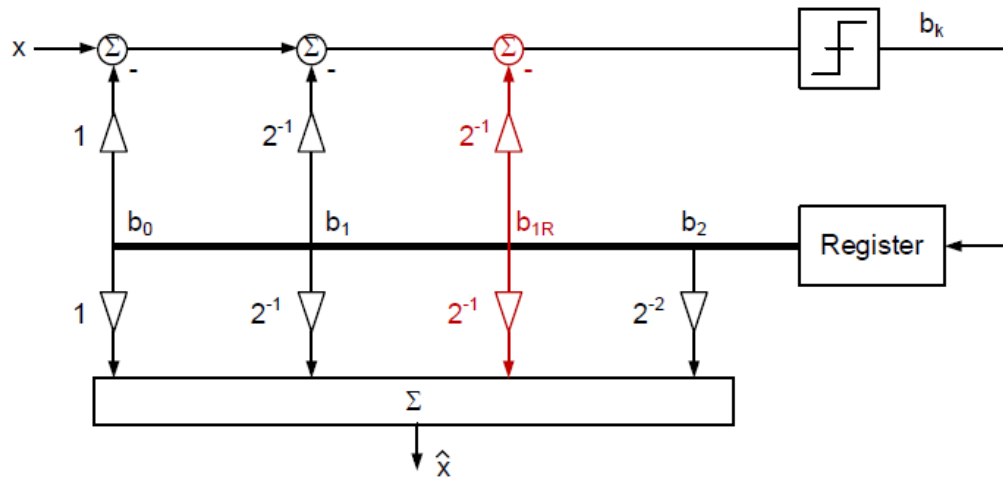


$$\beta < 2 \quad \sum_{i=1}^{B-1} \beta^{-i} > 1 \quad (\text{Key property})$$

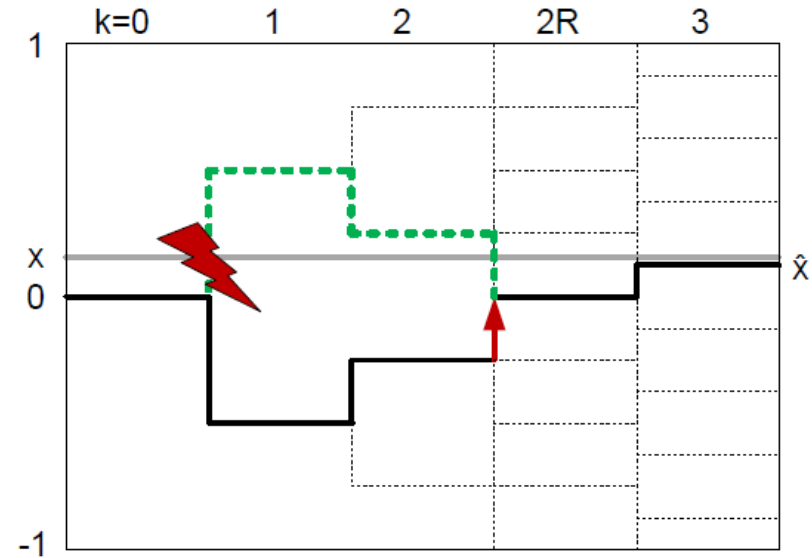


- $> N$ comparison for N bit
- Error can be tolerated by the redundancy range
- Work for comparator noise, DAC incomplete settling, & offset mismatch
- Green line $\rightarrow 1,0,0,1 \rightarrow D_{\text{out}} = \beta^{-1} + \beta^{-4}$
- Black line $\rightarrow 0,1,1,1 \rightarrow D_{\text{out}} = \beta^{-2} + \beta^{-3} + \beta^{-4}$

With redundancy (redundant bit)



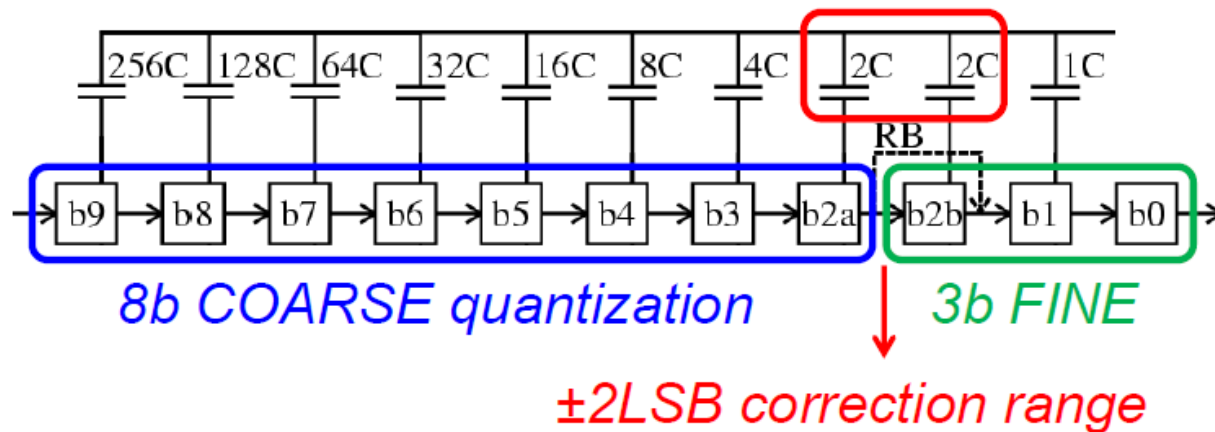
[C.-C. Liu, ISSCC 2010]



- $> N$ comparison for N bit
- Error can be tolerated before the redundant bit
- Green line: 1,0,0,1 $\rightarrow D_{\text{out}} = 2^{-1} + 2^{-3}$
- Black line: 0,1,1,1 $\rightarrow D_{\text{out}} = 2^{-2} + 2^{-2} + 2^{-3}$

An example for redundant bit

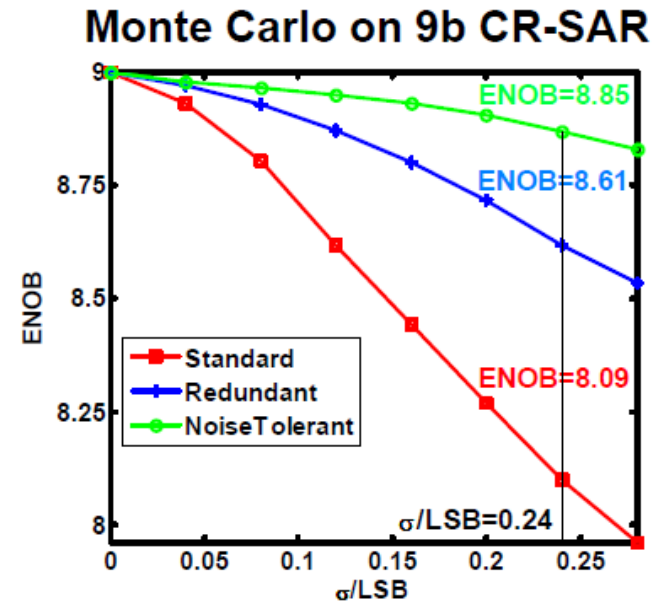
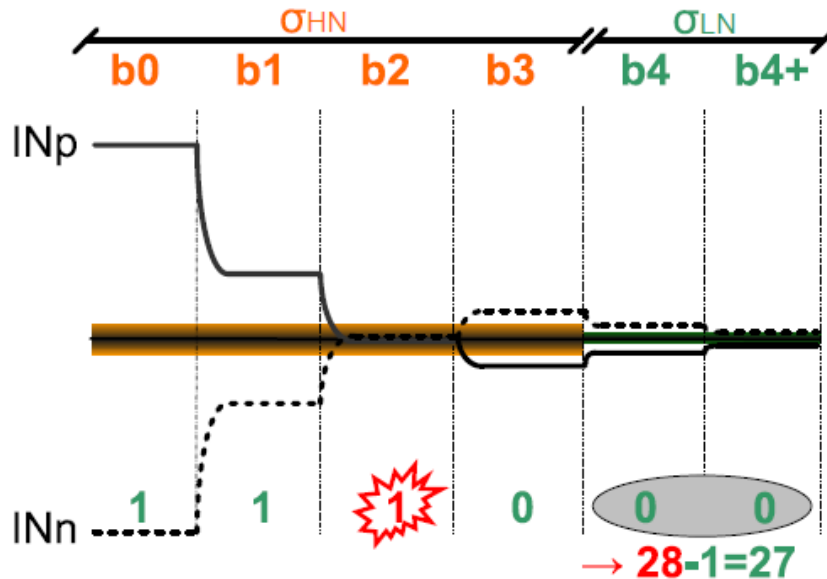
- 10bit SAR ADC $\rightarrow$ 11 cycles



- Coarse cycles: comparator in low-power mode
- Fine cycles: comparator in low-noise mode

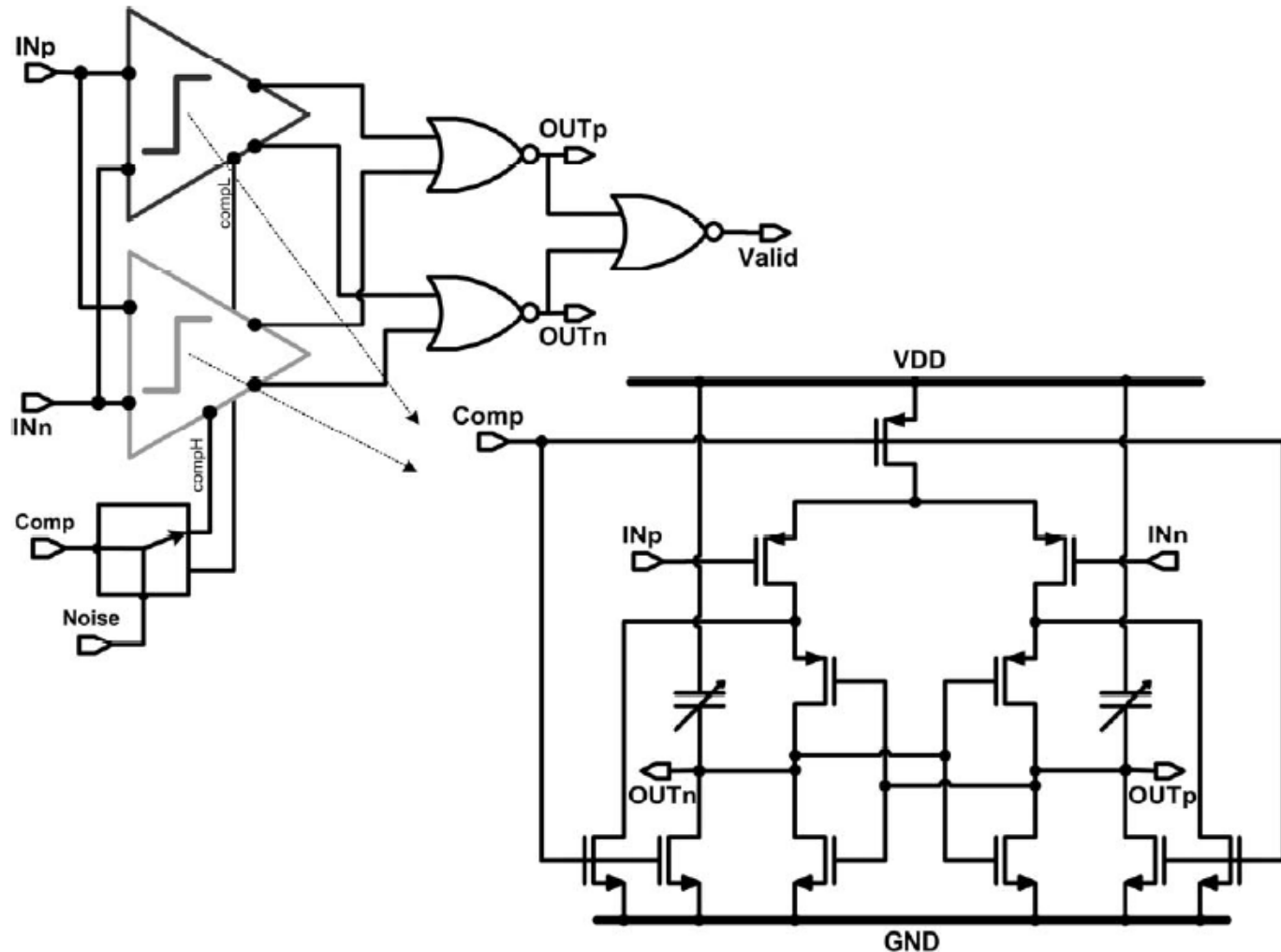
Comparator switching

1b redundancy + noise reconfigurable comparator



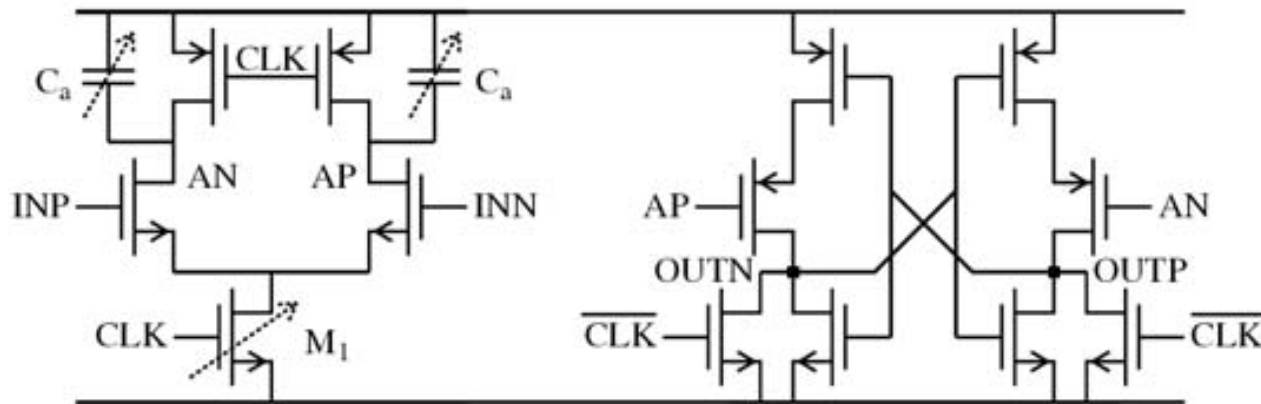
1. $(N-1)$ low power comparisons @ σ_{HN}
2. N and $N+1$ high power comparisons @ σ_{LN}

Comparator offset mismatch calibration



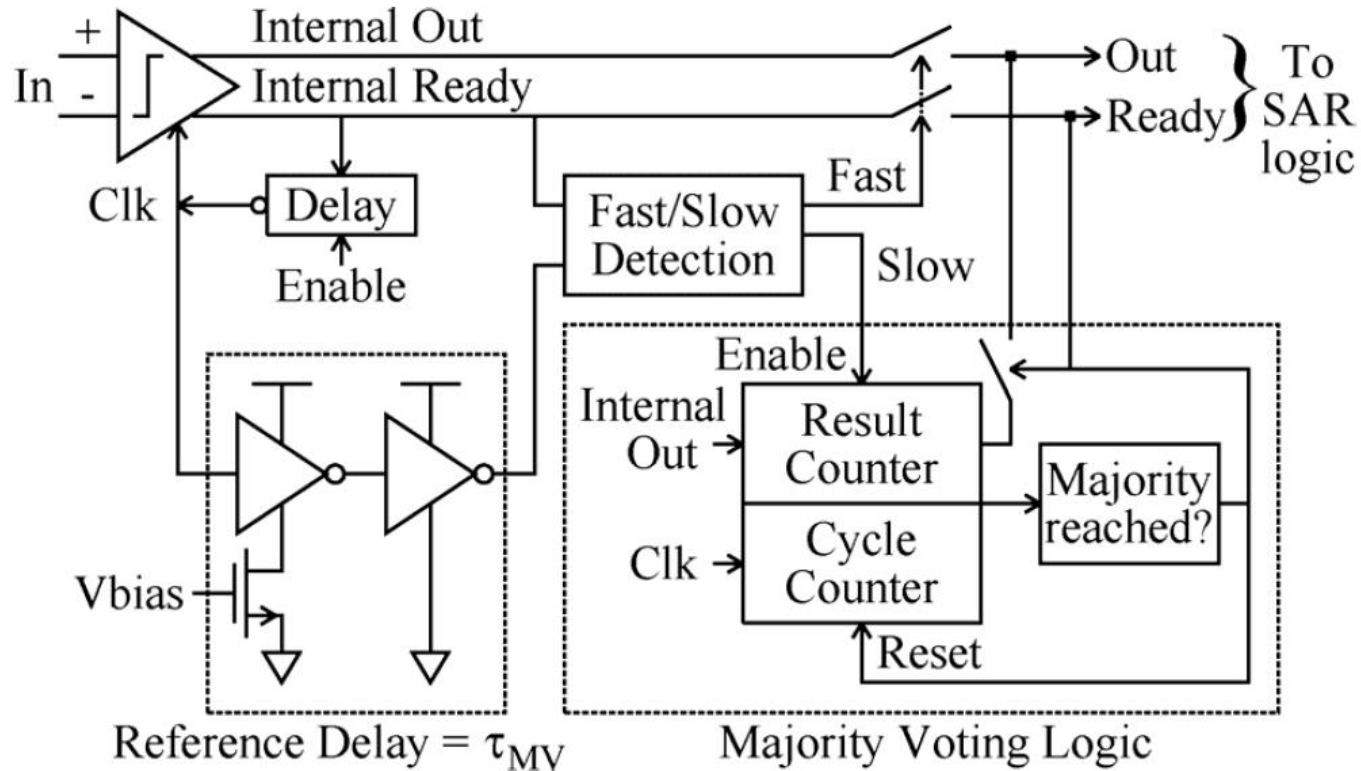
Comparator load switching

Share the same diff pair, and thus, offset mismatch is low



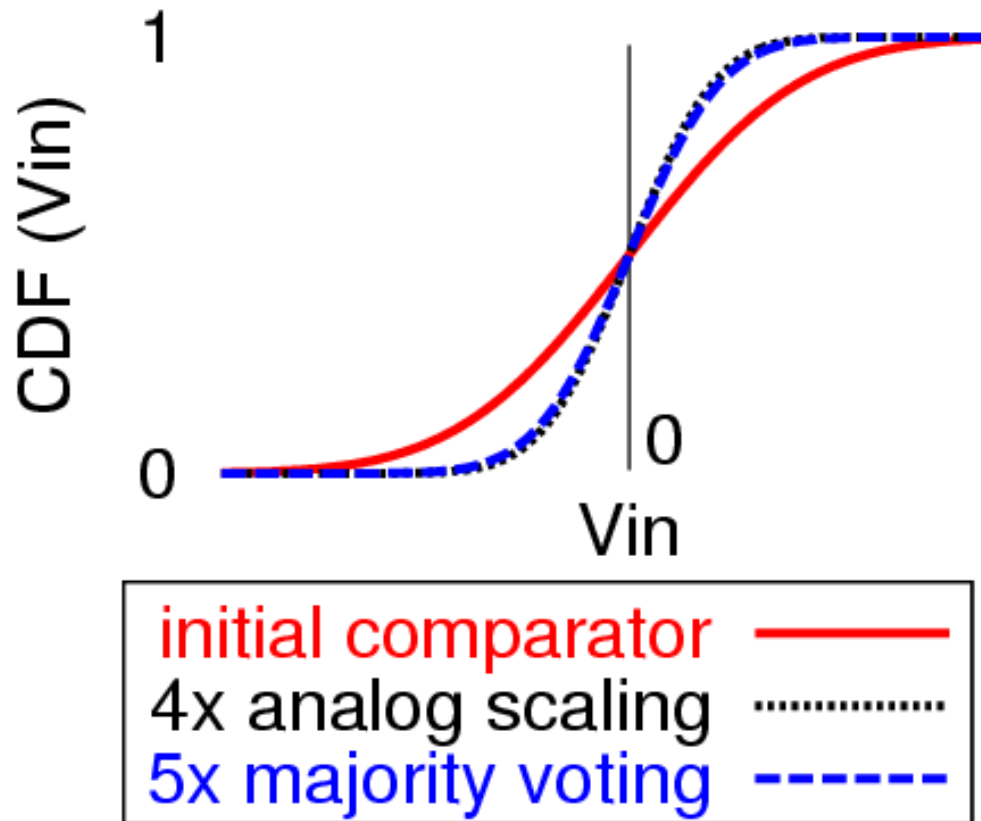
Accuracy setting	Energy per comparison	Input ref. noise
7b/8b	108fJ	0.24mVrms
9b	214fJ	0.18mVrms
10b	373fJ	0.12mVrms

Majority Voting



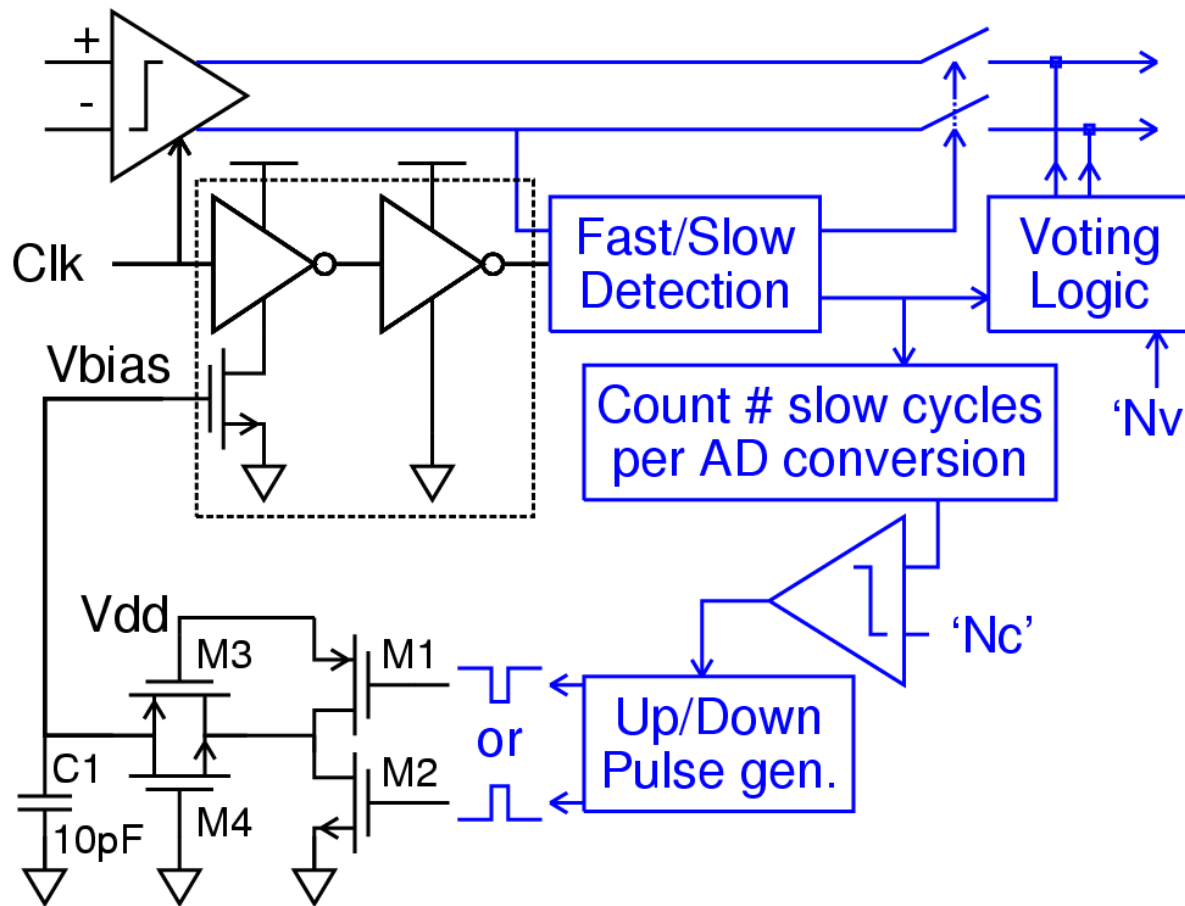
- Require additional circuits for critical moment detection
- Does not use all the available information

Majority Voting



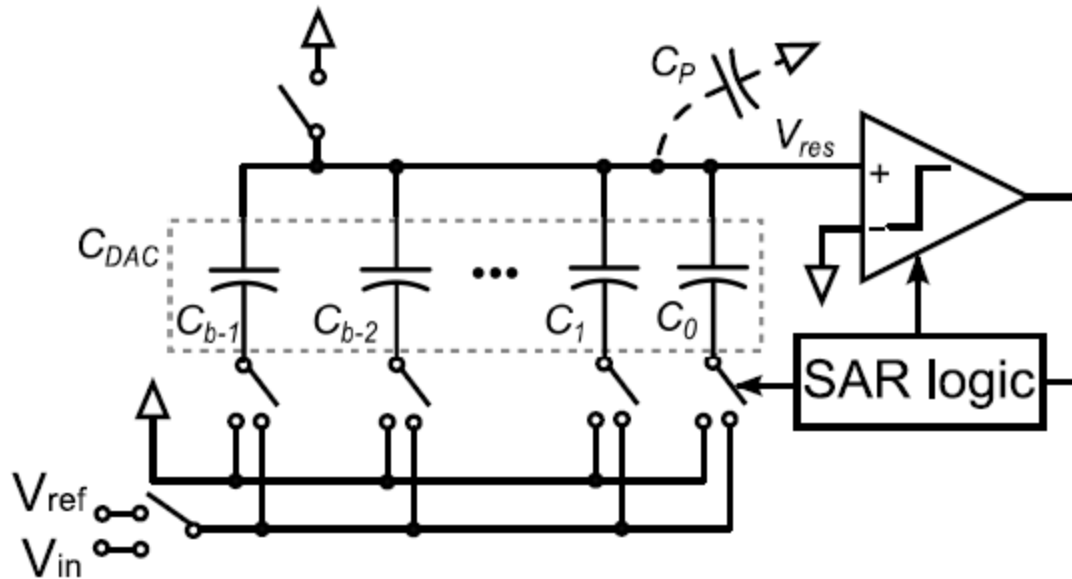
- For a 10-bit SAR ADC
 - 4x analog scaling $\rightarrow$ 4x total comparator power (10×4)
 - 5x majority voting $\rightarrow$ 1.4x total comparator power ($9 + 5$)

Background Regulation



- Background calibration for reference delay based on N_c monitoring
- N_c : number of target voting cycles

Basic idea for statistical estimation

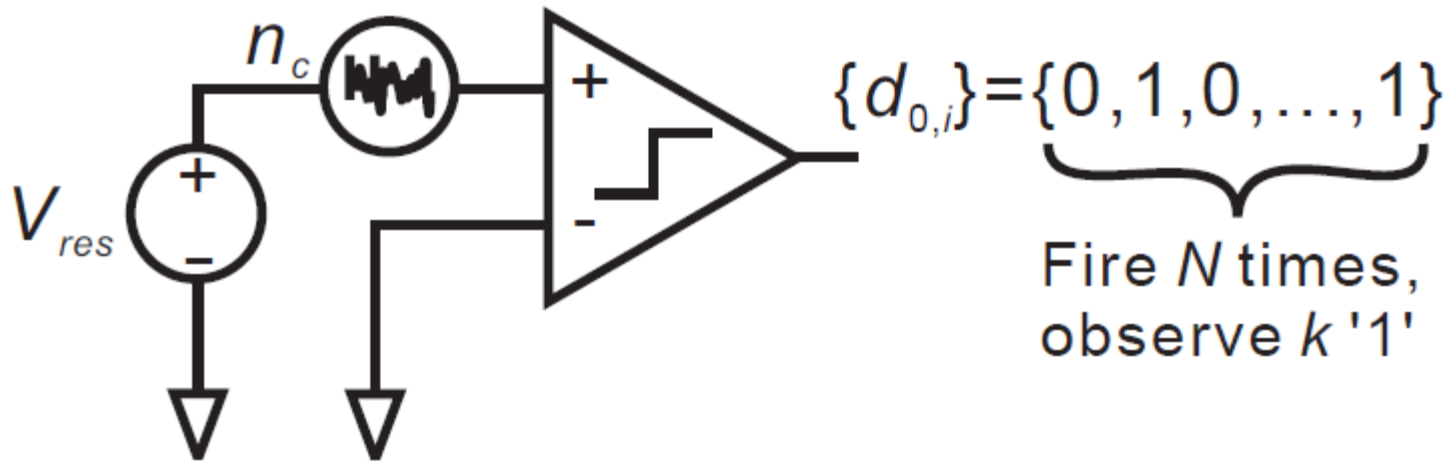


$$D_{out} = V_{in} + V_{res}$$

If we can estimate V_{res} using $\hat{V}_{res}$, we can increase ADC resolution by:

$$D_{out}^* = D_{out} - \hat{V}_{res} = V_{in} + (V_{res} - \hat{V}_{res})$$

Statistical estimation



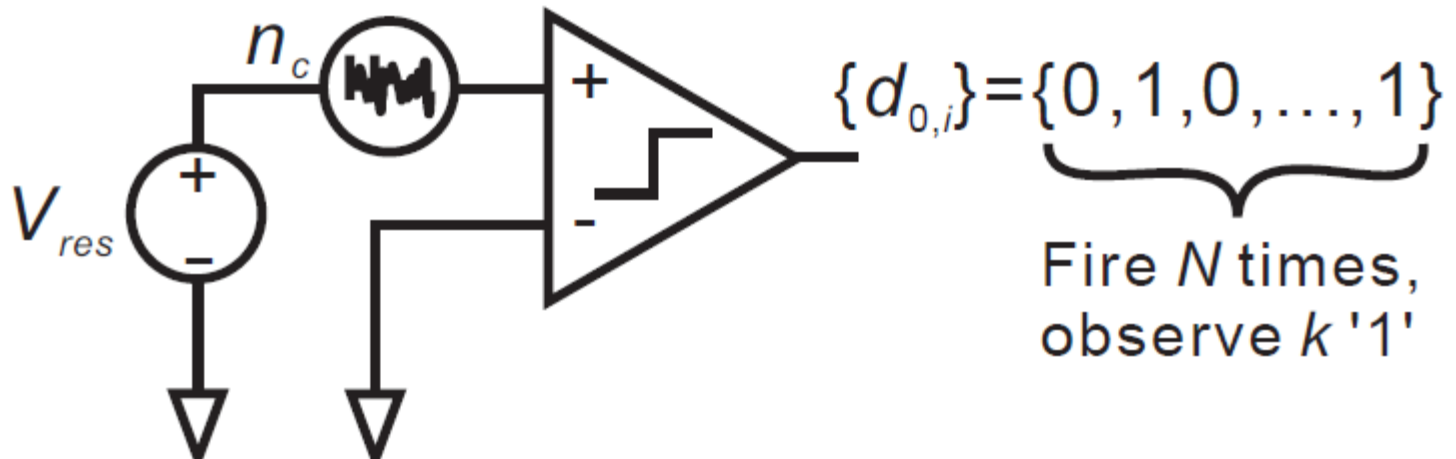
We can estimate V_{res} from k .
This is enabled by the noise!

Simple averaging

$$\hat{V}_{res} = \frac{2k - N}{N}$$

- Definition
 - $\hat{V}_{res} = +1$ LSB for $k = N$
 - $\hat{V}_{res} = -1$ LSB for $k = 0$
 - Linear interpolation in between
- Estimation error is large
 - Range limited to ± 1 LSB
 - With bias

Statistical estimation

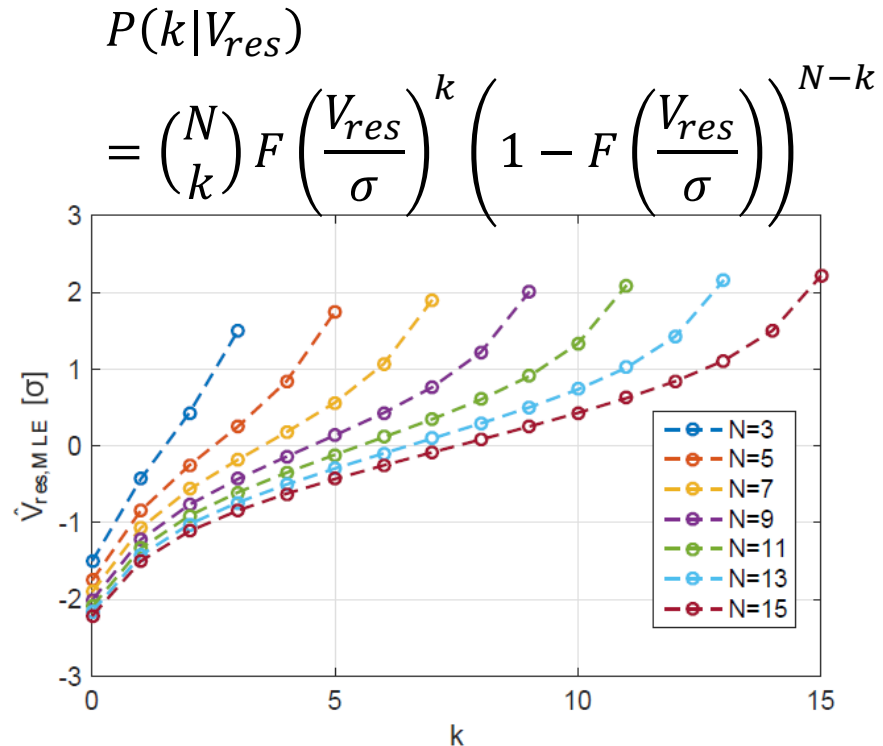


What information can we use to improve the estimation?

1. We can estimate the comparator noise power

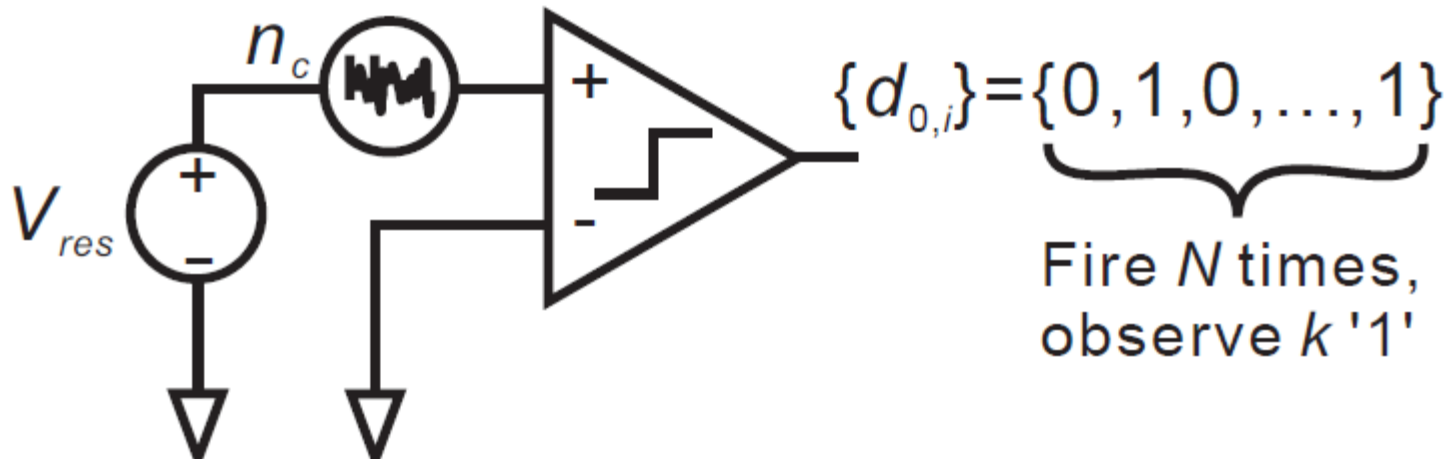
Maximum likelihood estimator (MLE)

Choose $\hat{V}_{res}$ that maximizes the probability of observing k '1' out of N trials:



- Ill-defined for $k = 0$ and $k = N$, need modification
- Sub-optimum

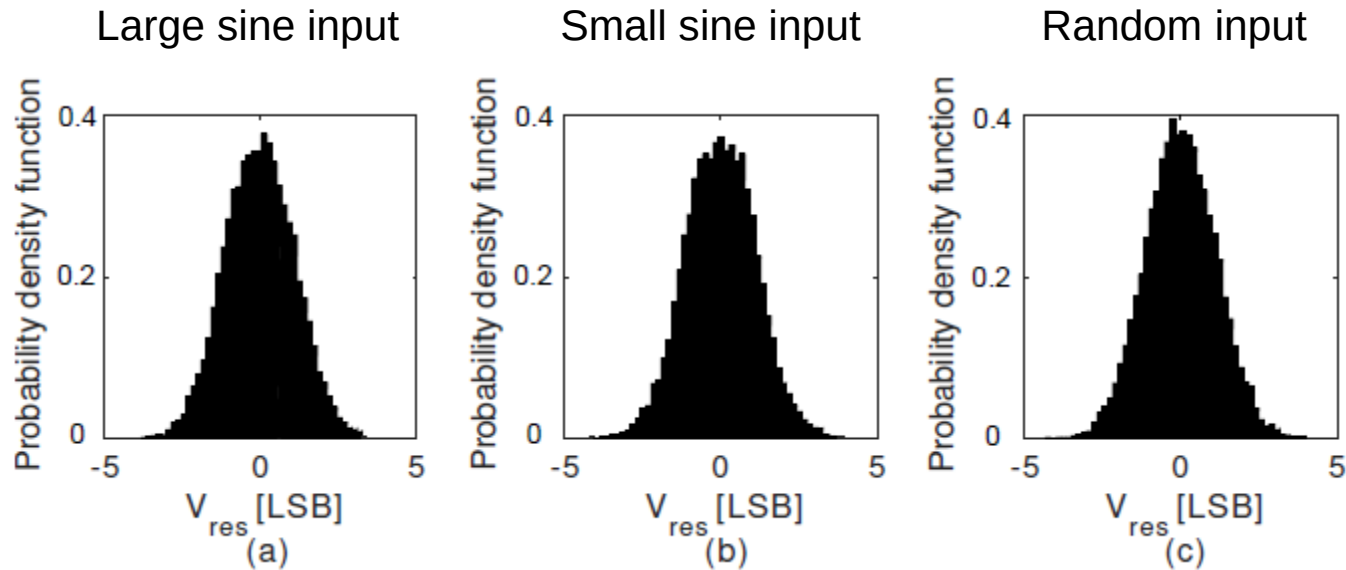
Statistical estimation



What information can we use to improve the estimation?

1. We can estimate the comparator noise power
2. We know the distribution of V_{res}

An additional piece of information

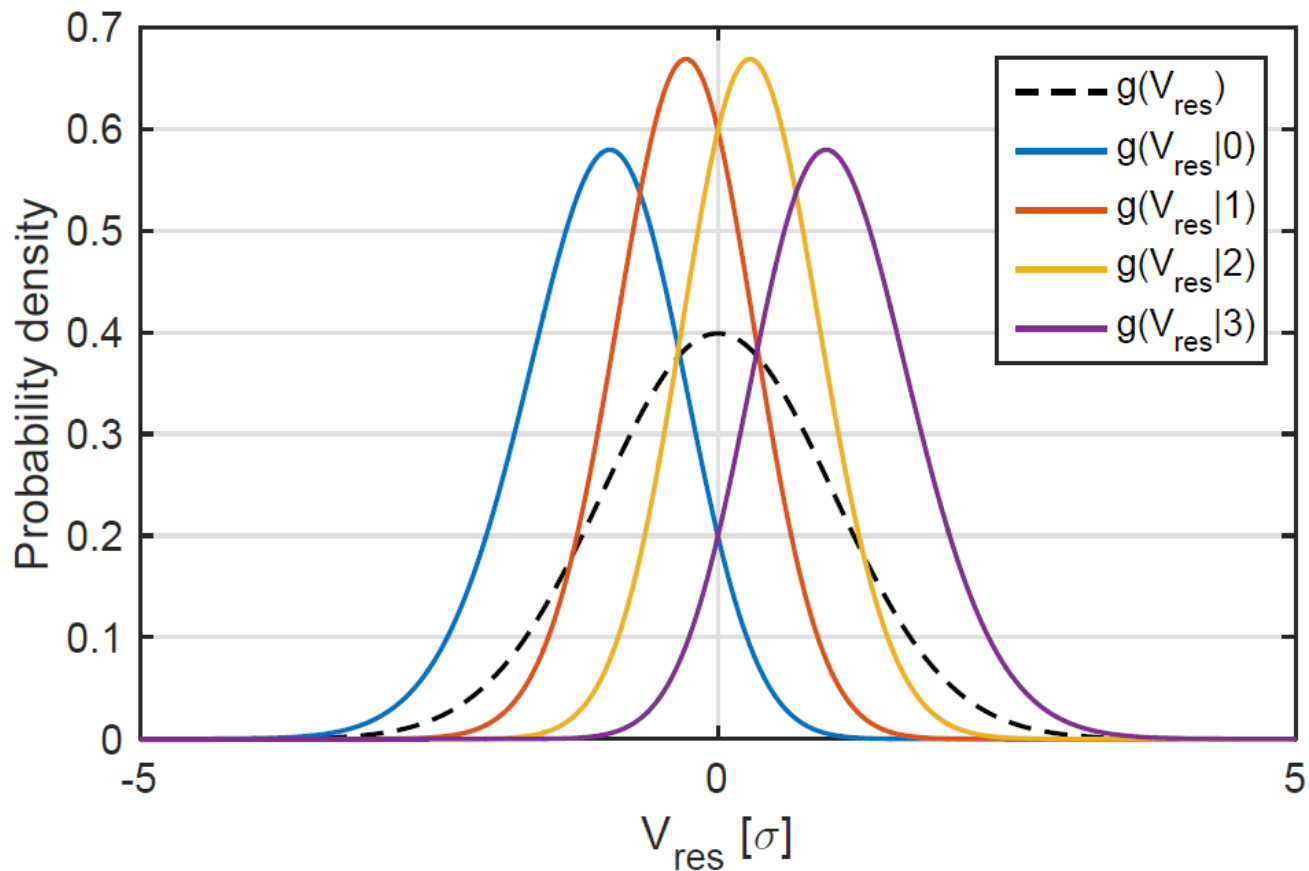


We know the distribution of V_{res} : $\text{norm}(0, \sigma)$

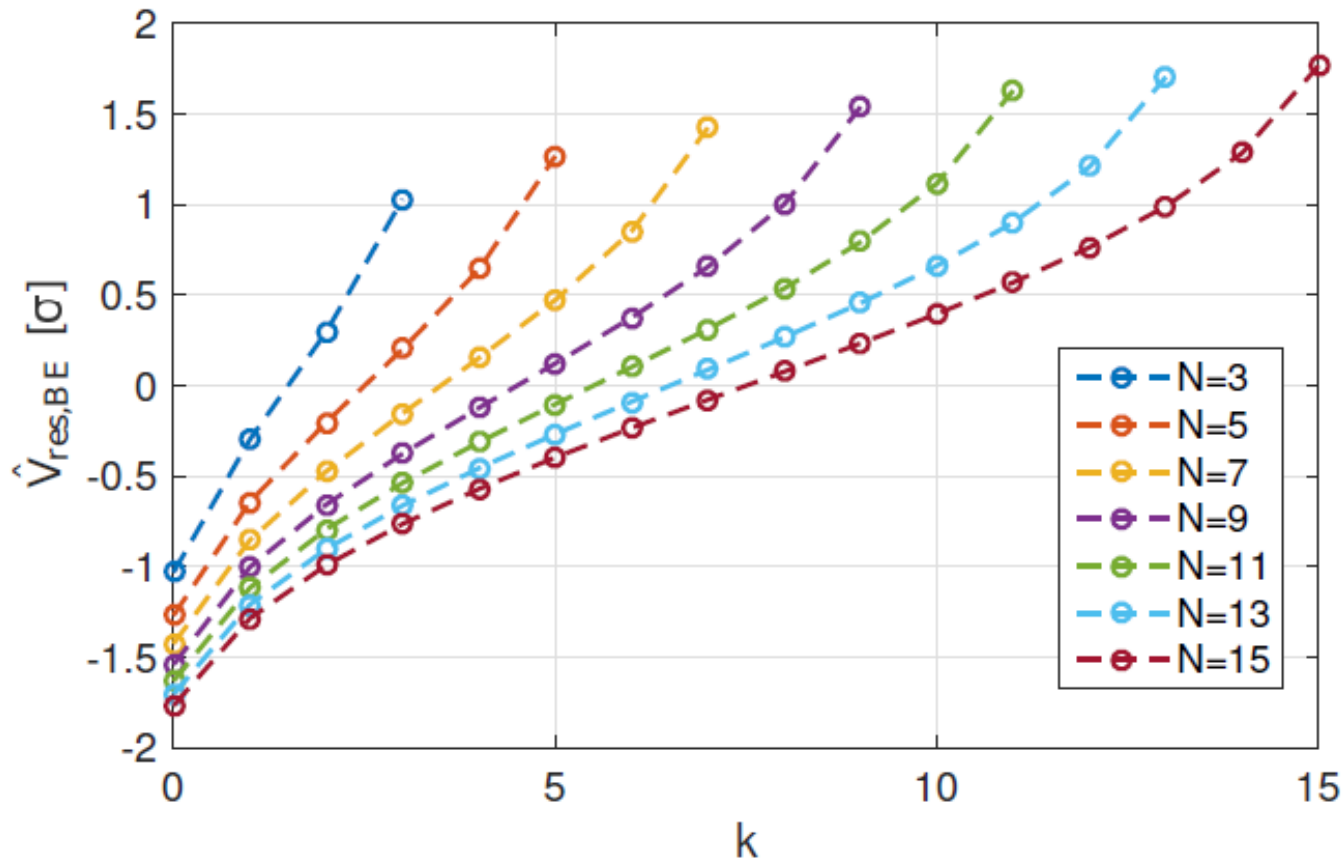
Bayes estimator

Choose posterior
mean

$$V_{res} = E(V_{res}|k) = \int_{-\infty}^{+\infty} V_{res} \cdot g(V_{res}|k) dV_{res}$$



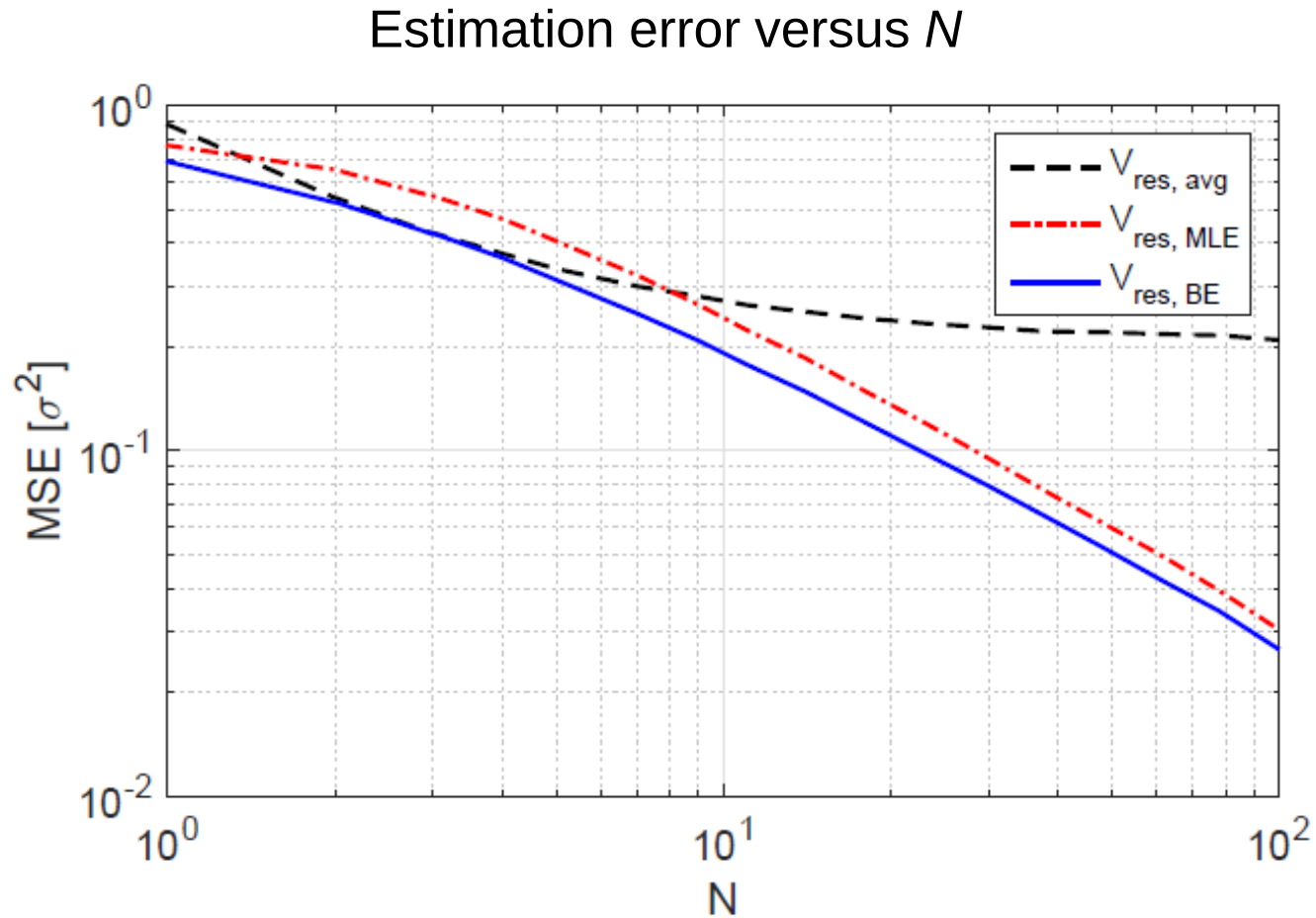
Bayes estimator



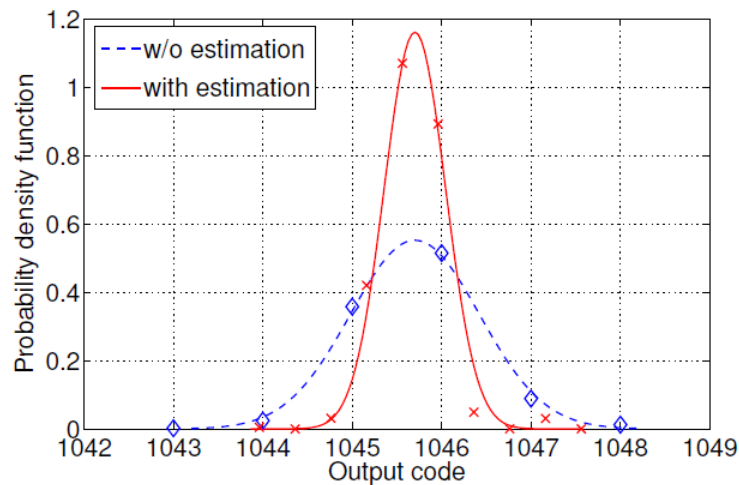
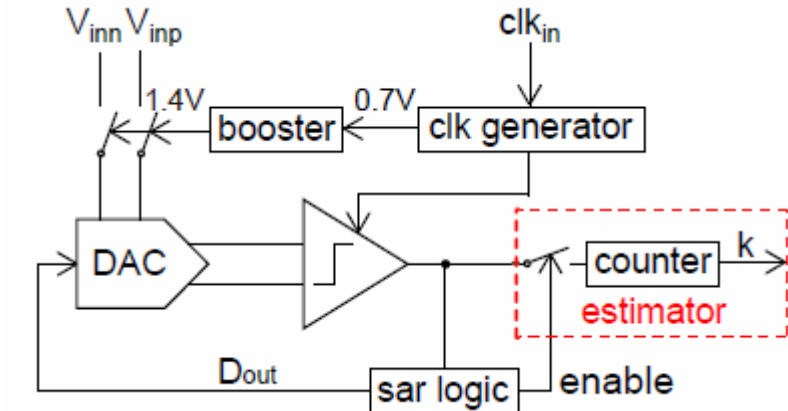
It achieves the lowest estimation error

$$MSE(k) = E \left[(V_{res} - \hat{V}_{res})^2 | k \right] = \left(\hat{V}_{res} - E(V_{res} | k) \right)^2 + Var(V_{res}^2 | k) \quad 54$$

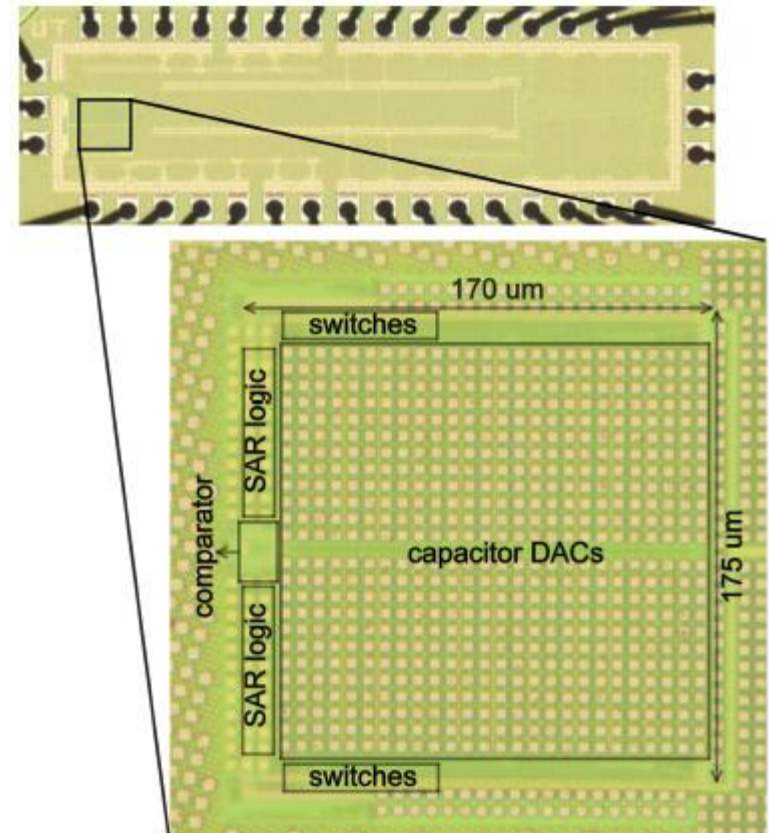
Estimator comparisons



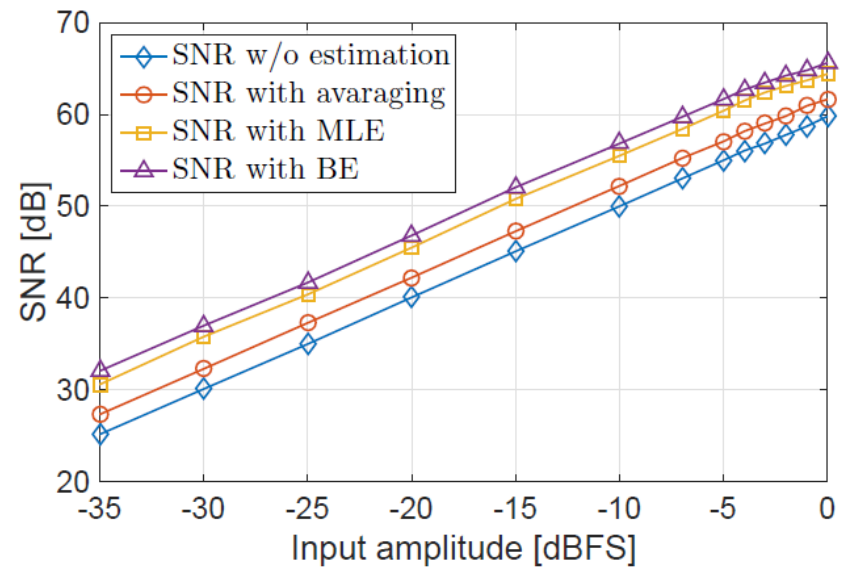
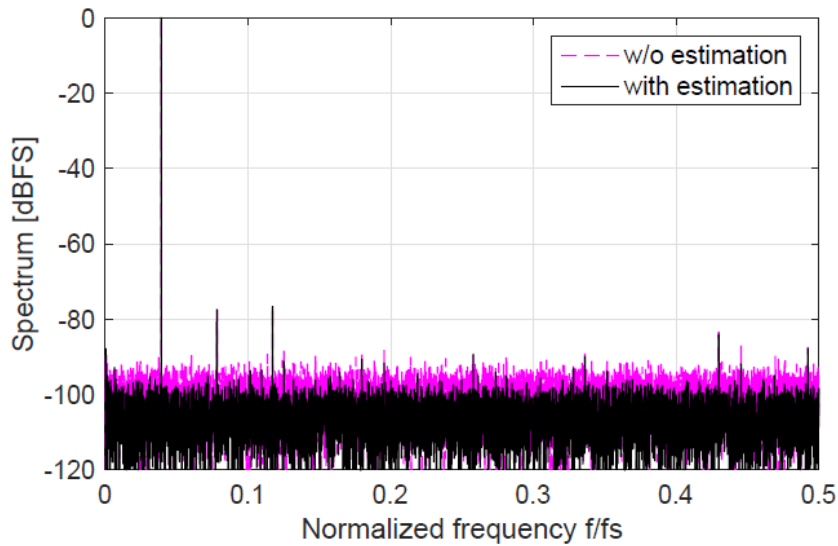
Prototype ADC Design



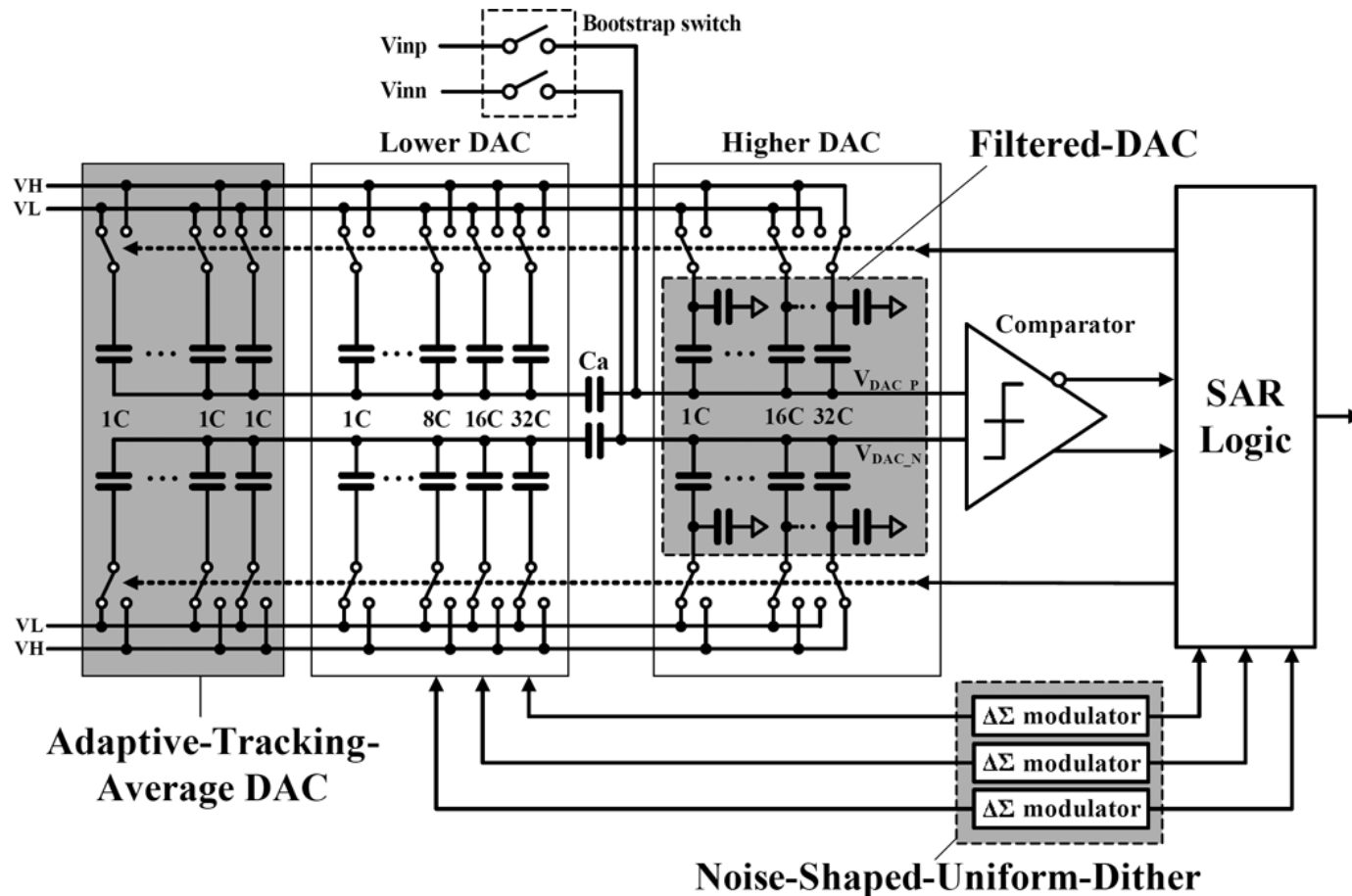
σ : 0.73 LSB $\rightarrow$ 0.33 LSB



Measurement results

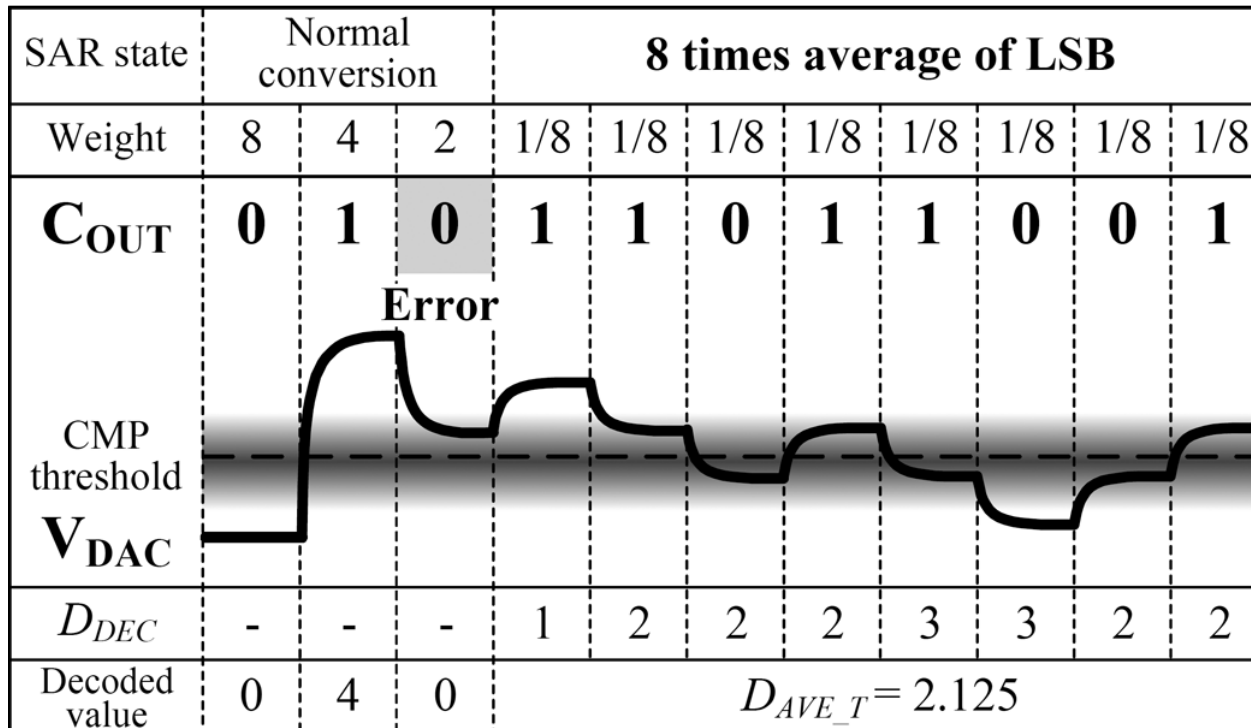


Adaptive Tracking Average



- Repeat the last LSB comparisons and also update DAC
- Perform averaging for the last few bits

Adaptive Tracking Average



$$D_{AVE_T} = \sum_{k=1}^M \frac{D_{DEC}[k]}{M}$$

- Simple averaging works well; can tolerate incomplete settling 😊
- Require additional LSB DAC cells 😞

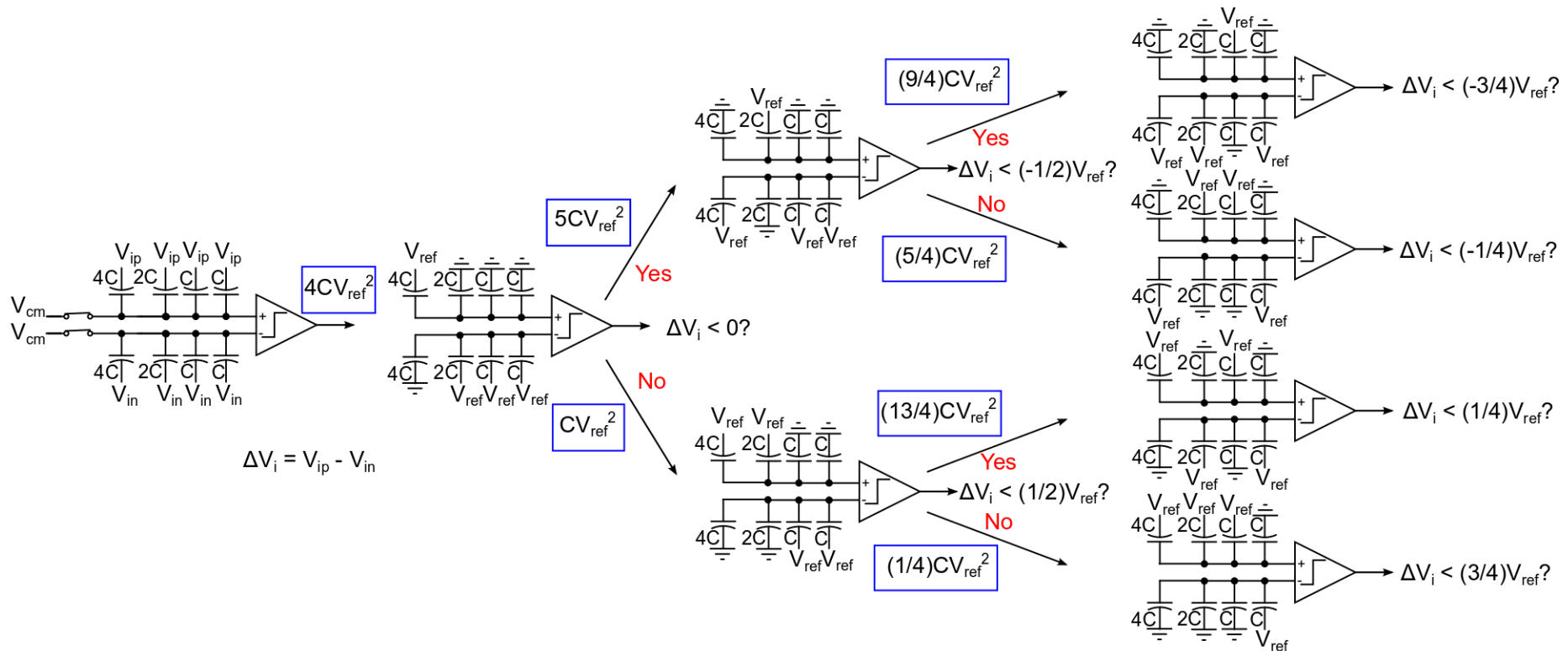
Low-noise low-power comparator design

- Transistor level (generally applicable):
 - Enlarge g_m/I_D
 - Enlarge preamplifier gain
 - Incomplete discharge of the integration capacitor
 - Reuse current (CMOS input)
- Architectural level (enabled by SAR's property & redundancy):
 - Comparator noise switching
 - Majority voting
 - Statistical estimation
 - Adaptive tracking average

Low-power DAC switching techniques

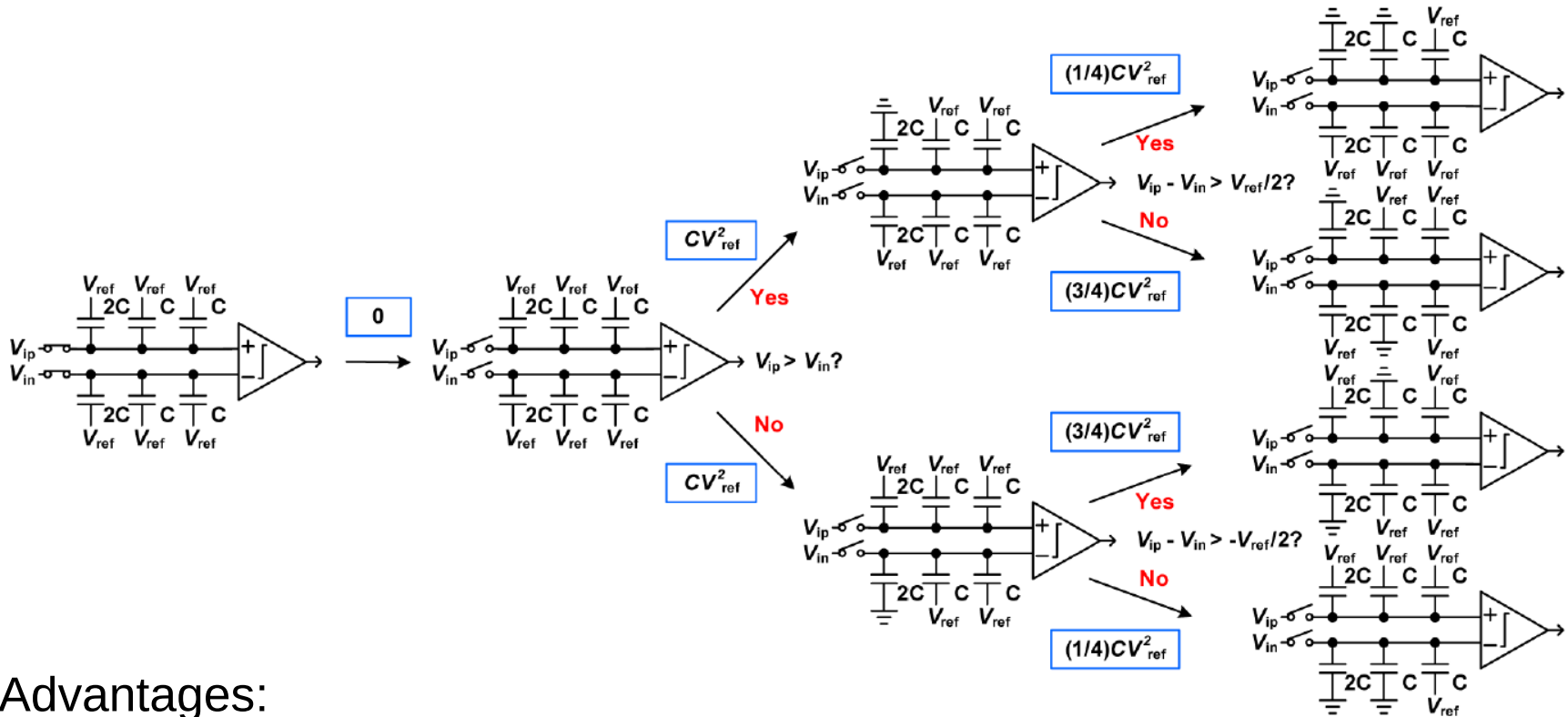
- Conventional switching technique
- Monotonic switching
- Vcm-based switching
- Bidirectional single-side switching
- Sub-ranging architecture

Conventional switching technique



- Need $16C$ for 3-b
- Maximum 4 switching events per bit
- Large DAC reference power

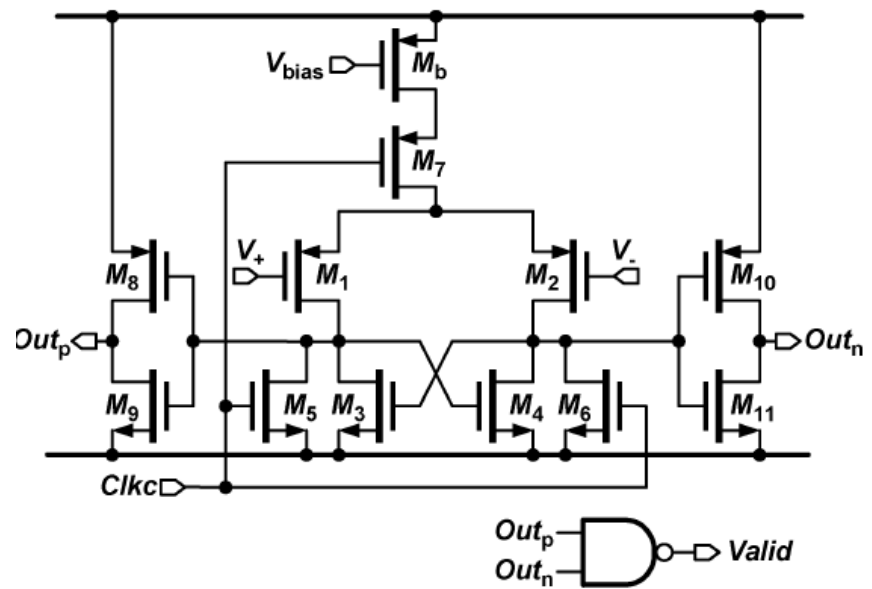
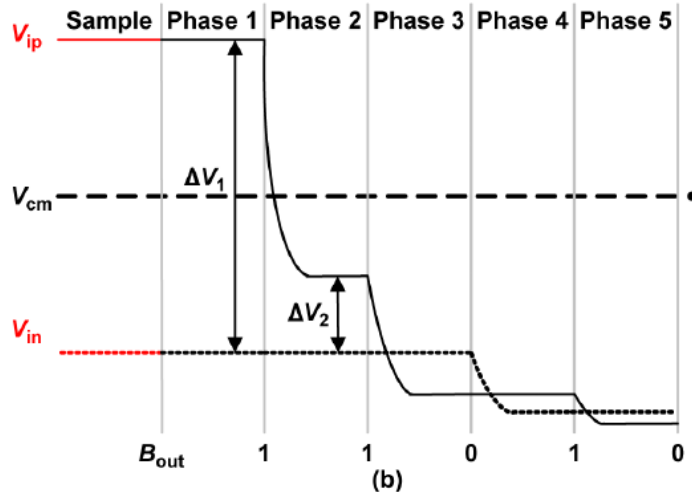
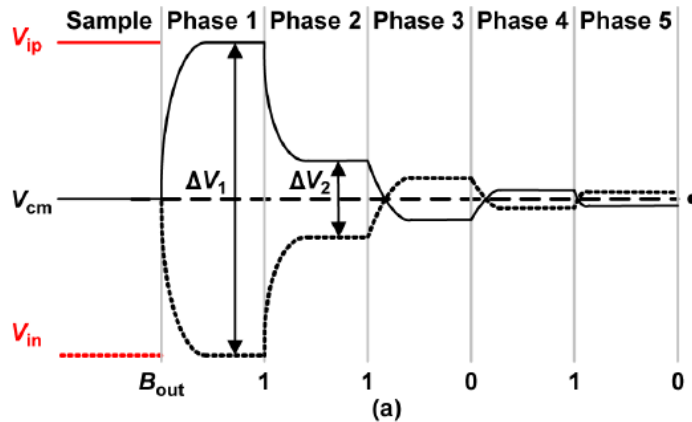
Monotonic switching technique



Advantages:

- 1) 81% reduction in DAC reference power
- 2) Only 1 switching every cycle

Common-mode variation

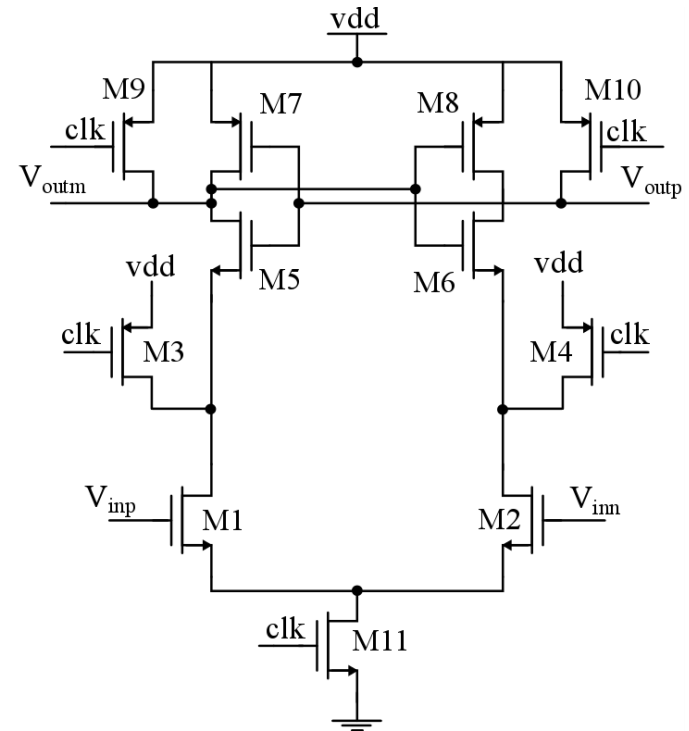
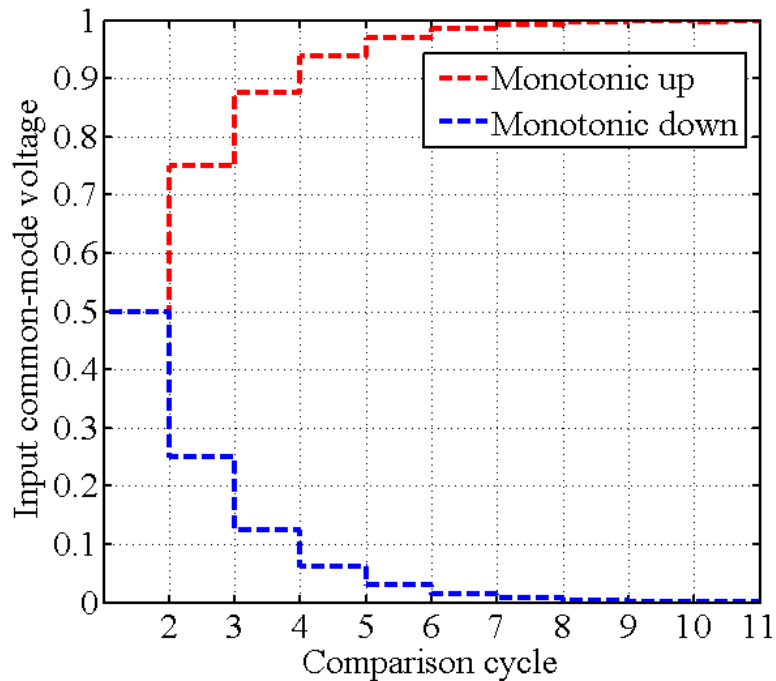


Disadvantages:

- 1) V_{cm} decreases to 0
- 2) PMOS input pair (smaller f_T)
- 3) Dissipate static power



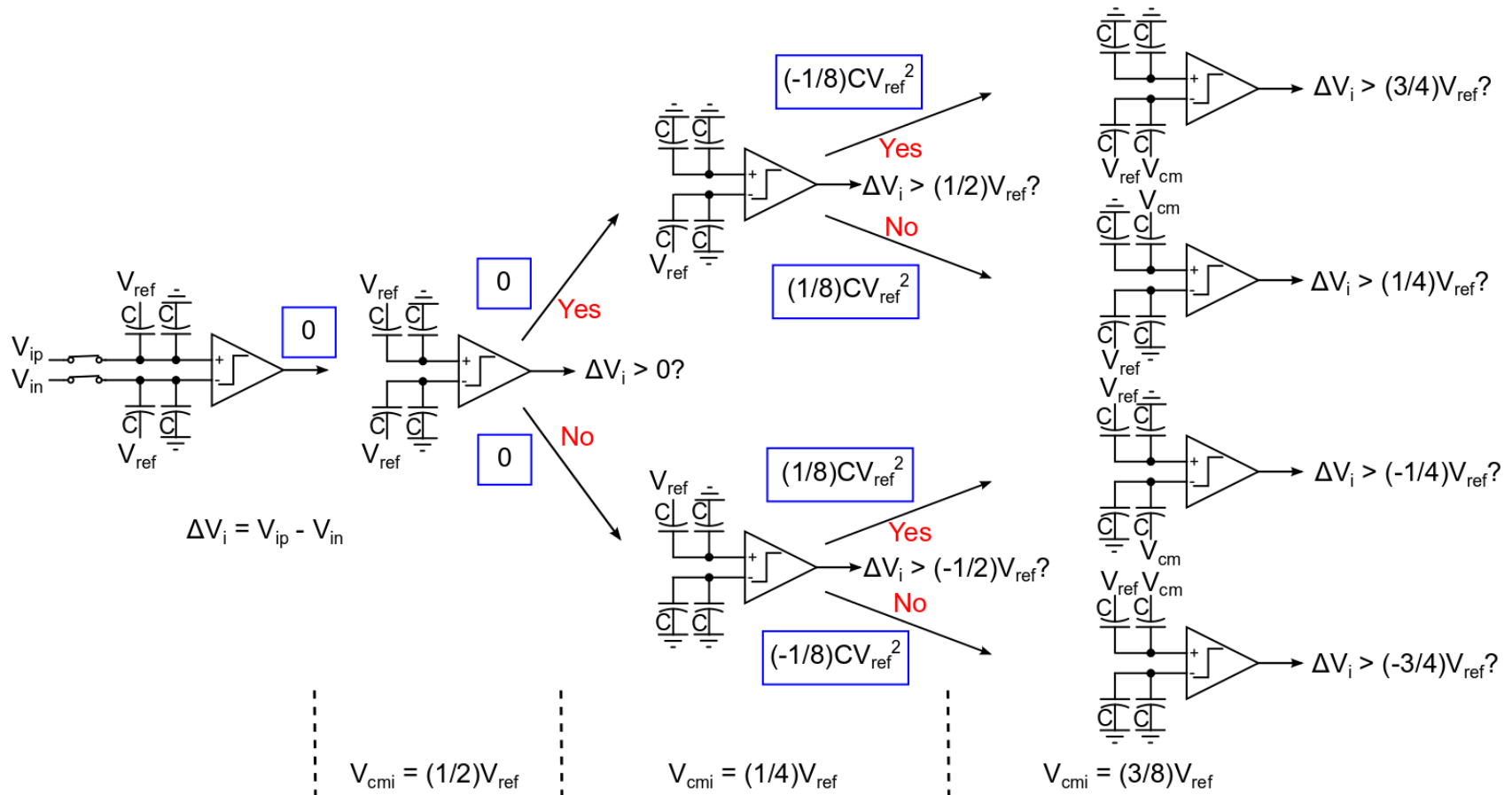
Monotonic switching up (2)



Disadvantages:

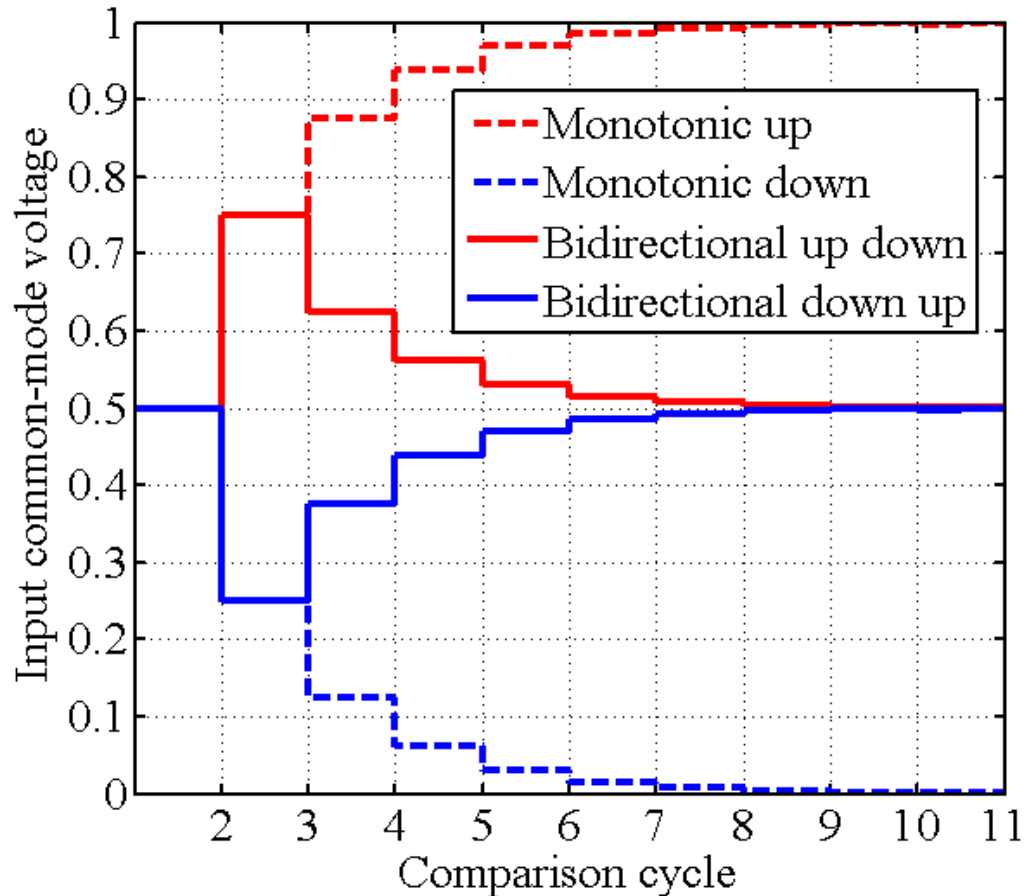
- 1) V_{cm} goes to V_{dd}
- 2) Large comparator offset
- 3) Large comparator noise

Bidirectional single-side switching



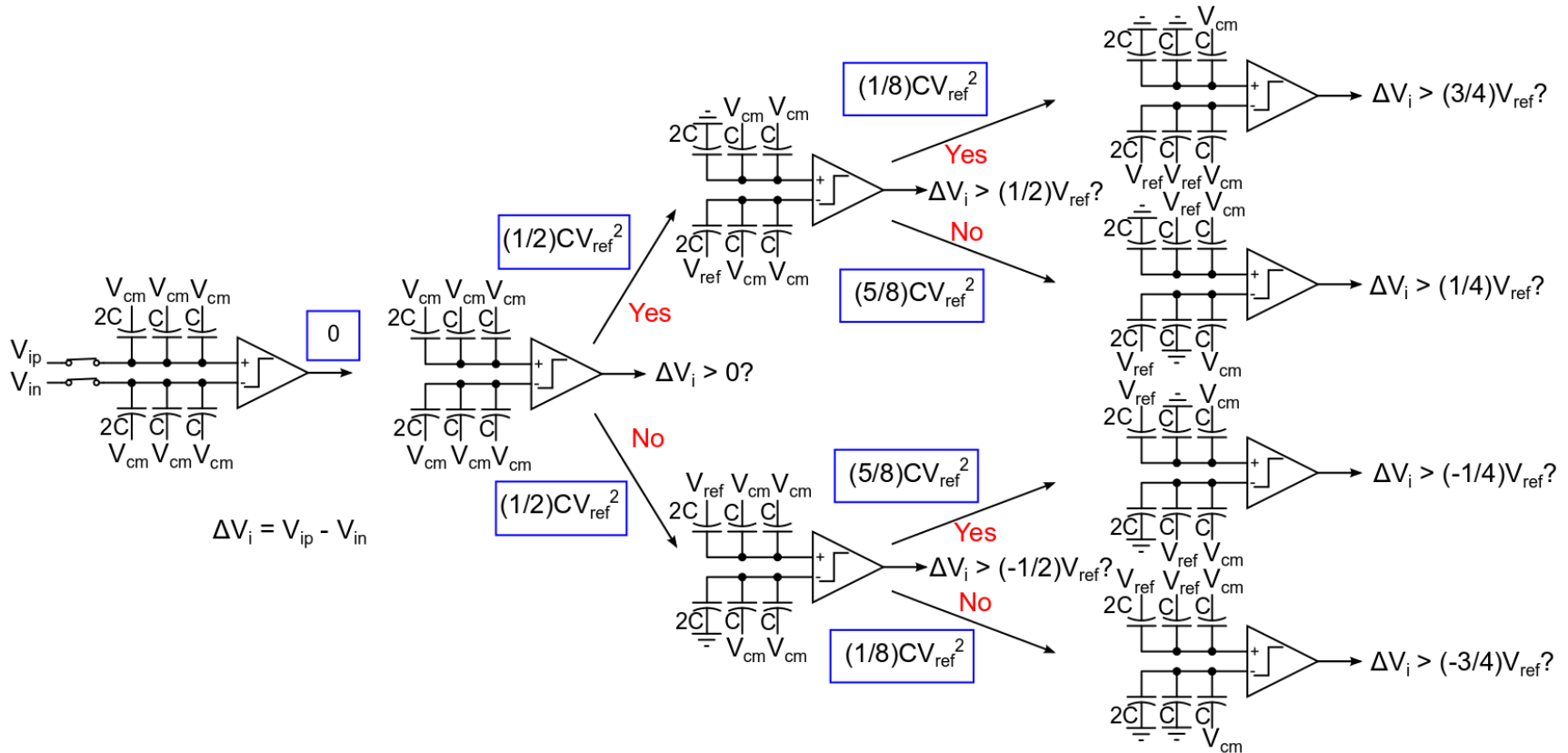
- First down switching and then up switching
- Reduced common-mode variation

Common-mode variation



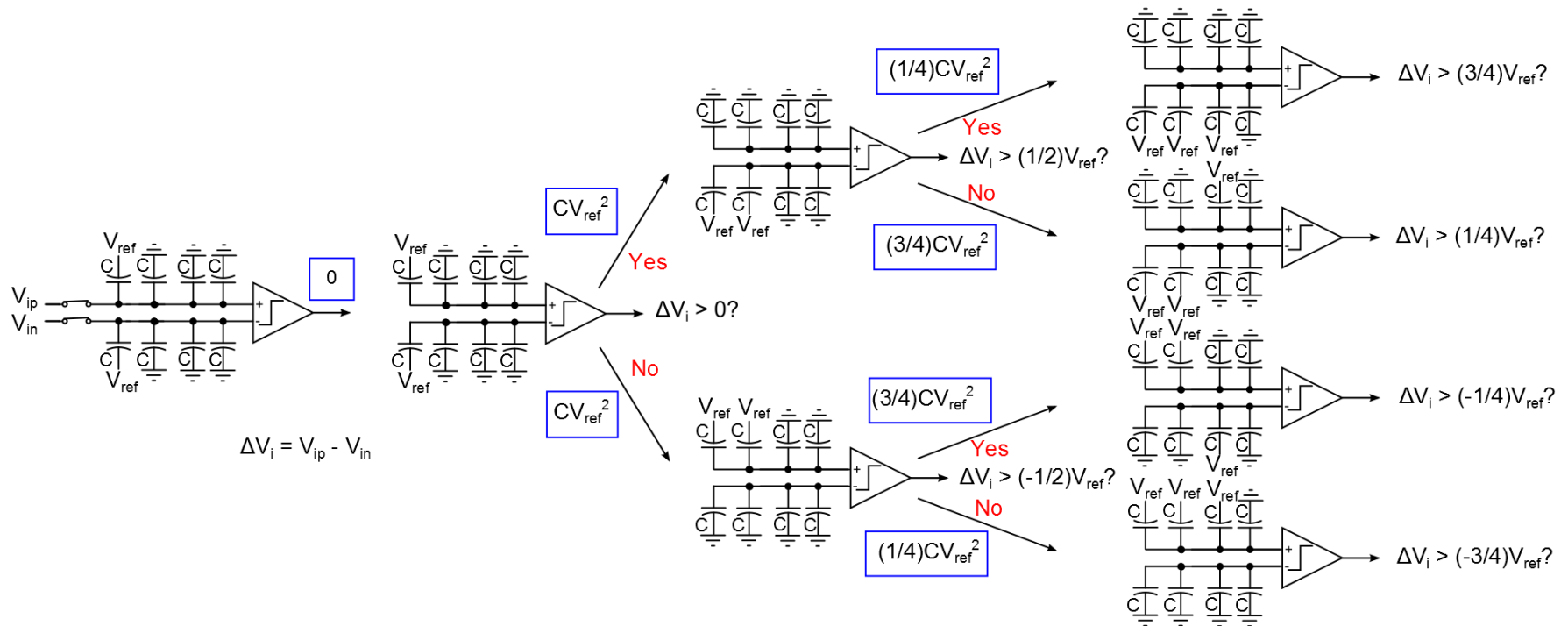
- Comparator input common-mode voltage settles to $V_{dd}/2$
- Standard comparator can be used; offset and noise are reduced

V_{cm} -based switching technique



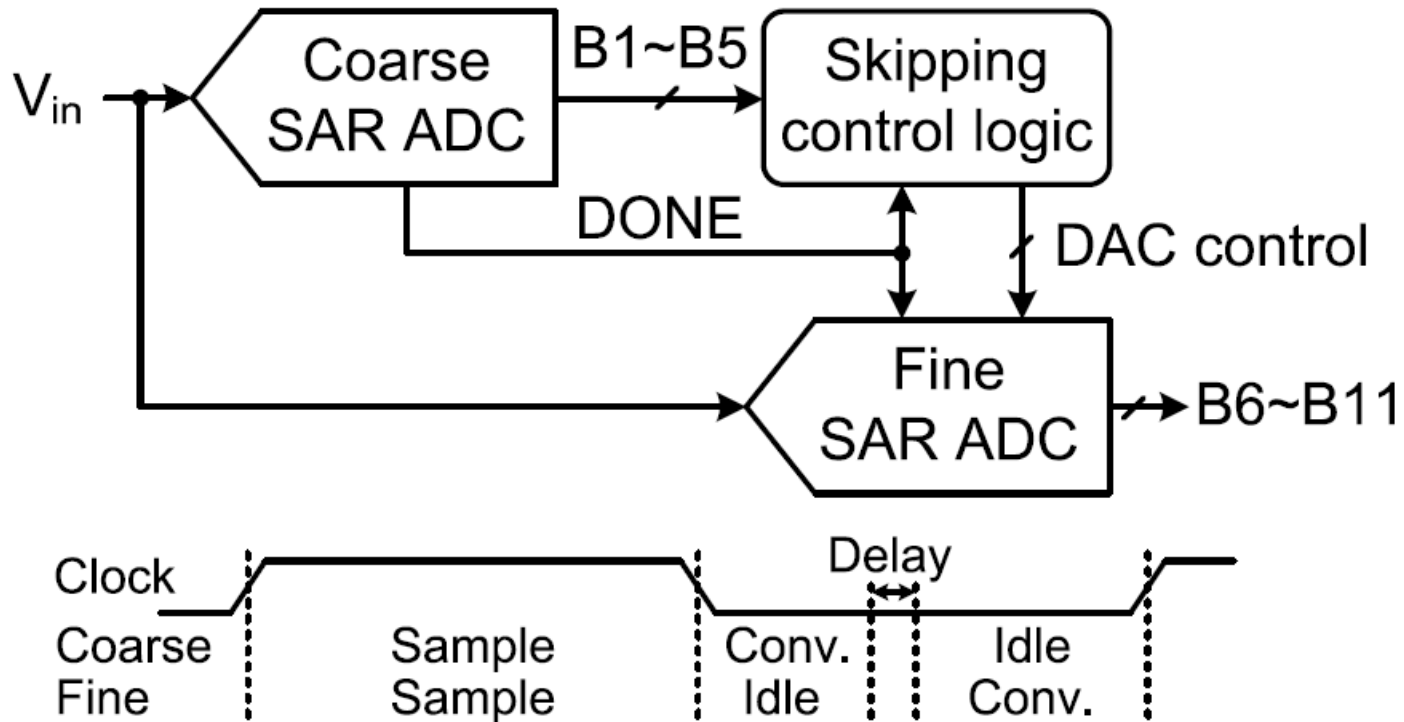
- Constant V_{cm} ; 88% energy saving
- Need low-impedance V_{cm} reference; increased switch driving power

Split MSB capacitor



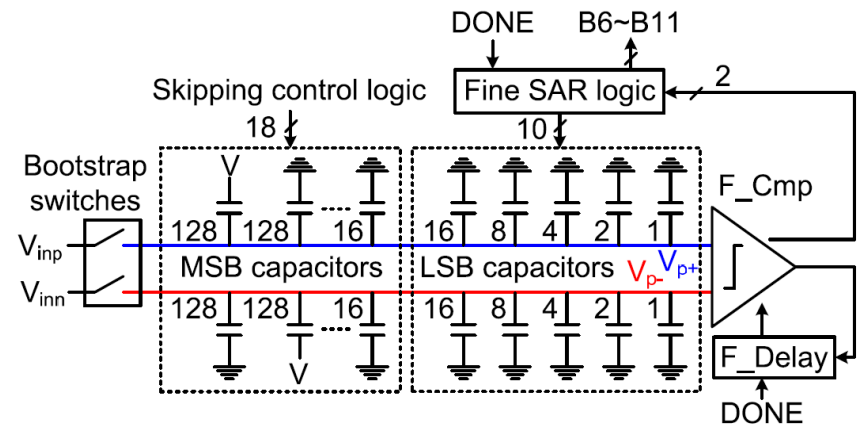
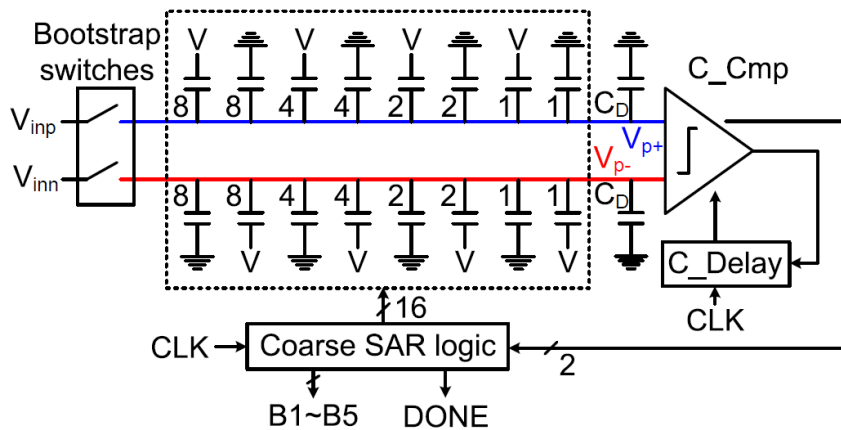
- No need for V_{cm} at the cost of increased switching power
- Additional capacitor mismatch

Saving Switching Energy Using Sub-ranging (1)



- Determine MSB bits in a coarse DAC
- Switch MSBs of fine DAC at the same time

Saving Switching Energy Using Sub-ranging (2)



SAR logic power reduction

- Reduce the number of DFFs
- Replace DFF by latch
- Use dynamic logic
- Use asynchronous logic
 - Remove the need for high-frequency clock

Re-cap

- SAR ADC noise analysis
 - kT/C noise
 - DAC noise
 - Comparator noise
- Comparator noise reduction techniques
 - Circuit level
 - Architectural level
- Low power DAC switching techniques